

Basics of electronics and components

4. Semiconductor Physics and Diode Characteristics

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Greek letters

Lower case greek letters & commonly used capitals

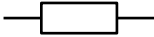
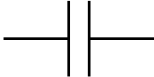
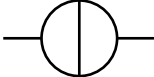
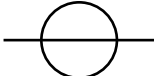
α	alpha
β	beta
$\gamma \Gamma$	gamma
$\delta \Delta$	delta
$\epsilon (\varepsilon)$	epsilon ('varepsilon')
ζ	zeta
η	eta
$\theta \Theta$	theta

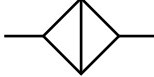
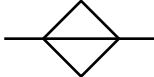
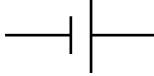
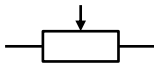
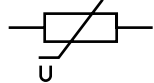
ι	iota
κ	kappa
$\lambda \Lambda$	lambda
μ	mu
ν	nu
$\xi \Xi$	xi
$\omicron$	omicron
$\pi \Pi$	pi

ρ	rho
$\sigma \Sigma$	sigma
τ	tau
$\upsilon \Upsilon$	upsilon
$\phi (\varphi) \Phi$	phi ('varphi')
χ	chi
$\psi \Psi$	psi
$\omega \Omega$	omega

Circuit elements

Basic passive and active elements

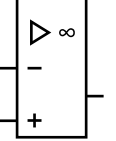
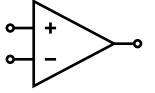
Type	EN-60617 symbol no.	
Resistor	04-01-01	
Capacitor (generic)	04-02-01	
Inductance (generic)	-	
Ground (generic)	02-15-01	
Current source	02-16-01	
Voltage source	02-16-02	

Type	EN-60617 symbol no.	
Dependent current source	-	
Dependent voltage source	-	
Battery cell	05-03-06	
Resistor (changeable)	04-01-03	
Potentiometer	04-01-07	
Varistor	04-01-04	

Circuit elements

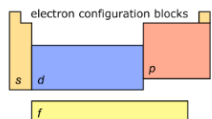
Active and passive semiconductor devices

Type	EN-60617 symbol no.	
Diode (generic)	05-03-01	
Schottky diode	05-02-04 + 05-03-01	
Zener diode	05-03-06	
LED	05-03-02	
Photo-diode	05-06-02	

Type	EN-60617 symbol no.	
Operational amplifier	Part 13	
Operational amplifier	(old symbol)	

The periodic table of the elements

group 1																		18
period 1	1																	2
	1.00794 1.0125 H Hydrogen 1s ¹																	4.002602 2.073 He Helium 1s ²
2	6.941 5202 0.98 Li Lithium 1s ² 2s ¹	9.012182 8965 1.57 Be Beryllium 1s ² 2s ²																10.1797 20.907 Ne Neon 1s ² 2s ² 2p ⁶
3	22.98976 4958 0.93 Na Sodium [Ne] 3s ¹	24.3050 7327 1.31 Mg Magnesium [Ne] 3s ²																39.948 15306 Ar Argon [Ne] 3s ² 3p ⁶
4	39.0983 4188 0.82 K Potassium [Ar] 4s ¹	40.078 5898 1.00 Ca Calcium [Ar] 4s ²	44.95591 8331 1.36 Sc Scandium [Ar] 3d ¹ 4s ²	47.867 5898 1.54 Ti Titanium [Ar] 3d ² 4s ²	50.9415 6009 1.03 V Vanadium [Ar] 3d ³ 4s ²	51.9962 6538 1.66 Cr Chromium [Ar] 3d ⁵ 4s ¹	54.93804 7113 1.55 Mn Manganese [Ar] 3d ⁵ 4s ²	55.845 7625 1.83 Fe Iron [Ar] 3d ⁶ 4s ²	58.93319 7604 1.91 Co Cobalt [Ar] 3d ⁷ 4s ²	58.93319 7604 1.91 Ni Nickel [Ar] 3d ⁸ 4s ²	63.546 7455 1.90 Cu Copper [Ar] 3d ¹⁰ 4s ¹	65.38 7920 1.69 Zn Zinc [Ar] 3d ¹⁰ 4s ²	68.723 7920 1.81 Ga Gallium [Ar] 3d ¹⁰ 4s ² 4p ¹	72.64 7602 2.01 Ge Germanium [Ar] 3d ¹⁰ 4s ² 4p ²	74.92160 8340 2.05 As Arsenic [Ar] 3d ¹⁰ 4s ² 4p ³	78.96 8340 2.55 Se Selenium [Ar] 3d ¹⁰ 4s ² 4p ⁴	79.904 8340 2.55 Br Bromine [Ar] 3d ¹⁰ 4s ² 4p ⁵	83.798 11504 3.00 Kr Krypton [Ar] 3d ¹⁰ 4s ² 4p ⁶
5	85.4678 4030 0.82 Rb Rubidium [Kr] 5s ¹	87.62 5495 0.95 Sr Strontium [Kr] 5s ²	88.90585 8000 1.22 Y Yttrium [Kr] 4d ¹ 5s ²	91.224 6401 1.33 Zr Zirconium [Kr] 4d ² 5s ²	92.90638 6521 1.60 Nb Niobium [Kr] 4d ⁴ 5s ¹	95.96 6843 2.16 Mo Molybdenum [Kr] 4d ⁵ 5s ¹	(98) 7020 1.90 Tc Technetium [Kr] 4d ⁵ 5s ²	101.07 7102 2.20 Ru Ruthenium [Kr] 4d ⁷ 5s ¹	102.9055 7197 2.28 Rh Rhodium [Kr] 4d ⁸ 5s ¹	106.42 8044 2.20 Pd Palladium [Kr] 4d ¹⁰	107.8682 7310 1.93 Ag Silver [Kr] 4d ¹⁰ 5s ¹	112.414 8678 1.96 Cd Cadmium [Kr] 4d ¹⁰ 5s ²	114.818 5883 1.78 In Indium [Kr] 4d ¹⁰ 5s ² 5p ¹	118.710 7066 1.96 Sn Tin [Kr] 4d ¹⁰ 5s ² 5p ²	121.760 8340 2.05 Sb Antimony [Kr] 4d ¹⁰ 5s ² 5p ³	127.60 8693 2.52 Te Tellurium [Kr] 4d ¹⁰ 5s ² 5p ⁴	126.9044 10084 2.56 I Iodine [Kr] 4d ¹⁰ 5s ² 5p ⁵	131.293 11704 2.60 Xe Xenon [Kr] 4d ¹⁰ 5s ² 5p ⁶
6	132.9054 3757 0.79 Cs Cesium [Xe] 6s ¹	137.327 5019 0.89 Ba Barium [Xe] 6s ²	174.9668 5235 1.27 Lu Lutetium [Xe] 4f ¹⁴ 5d ¹ 6s ²	178.49 6965 1.30 Hf Hafnium [Xe] 4f ¹⁴ 5d ² 6s ²	180.9478 7912 1.30 Ta Tantalum [Xe] 4f ¹⁴ 5d ³ 6s ²	183.84 6965 1.30 W Tungsten [Xe] 4f ¹⁴ 5d ⁴ 6s ²	186.207 7030 2.36 Re Rhenium [Xe] 4f ¹⁴ 5d ⁵ 6s ²	190.23 7030 2.36 Os Osmium [Xe] 4f ¹⁴ 5d ⁶ 6s ²	192.217 7030 2.36 Ir Iridium [Xe] 4f ¹⁴ 5d ⁷ 6s ²	195.084 7030 2.36 Pt Platinum [Xe] 4f ¹⁴ 5d ⁹ 6s ¹	196.9665 8901 2.54 Au Gold [Xe] 4f ¹⁴ 5d ¹⁰ 6s ¹	200.59 10011 2.50 Hg Mercury [Xe] 4f ¹⁴ 5d ¹⁰ 6s ²	204.3833 5894 1.62 Tl Thallium [Xe] 4f ¹⁴ 5d ¹⁰ 6s ² 6p ¹	207.2 7156 2.33 Pb Lead [Xe] 4f ¹⁴ 5d ¹⁰ 6s ² 6p ²	208.9804 7156 2.33 Bi Bismuth [Xe] 4f ¹⁴ 5d ¹⁰ 6s ² 6p ³	(210) 7156 2.33 Po Polonium [Xe] 4f ¹⁴ 5d ¹⁰ 6s ² 6p ⁴	(210) 7156 2.33 At Astatine [Xe] 4f ¹⁴ 5d ¹⁰ 6s ² 6p ⁵	(220) 7156 2.33 Rn Radon [Xe] 4f ¹⁴ 5d ¹⁰ 6s ² 6p ⁶
7	(223) 3800 0.70 Fr Francium [Rn] 7s ¹	(226) 5093 0.88 Ra Radium [Rn] 7s ²	(262) 4700 1.03 Lr Lawrencium [Rn] 5f ¹⁴ 6d ¹ 7s ²	(261) 5800 1.04 Rf Rutherfordium [Rn] 5f ¹⁴ 6d ² 7s ²	(262) 5800 1.04 Db Dubnium [Rn] 5f ¹⁴ 6d ³ 7s ²	(266) 5800 1.04 Sg Seaborgium [Rn] 5f ¹⁴ 6d ⁴ 7s ²	(264) 5800 1.04 Bh Bohrium [Rn] 5f ¹⁴ 6d ⁵ 7s ²	(277) 5800 1.04 Hs Hassium [Rn] 5f ¹⁴ 6d ⁶ 7s ²	(268) 5800 1.04 Mt Meitnerium [Rn] 5f ¹⁴ 6d ⁷ 7s ²	(271) 5800 1.04 Ds Darmstadtium [Rn] 5f ¹⁴ 6d ⁸ 7s ²	(272) 5800 1.04 Rg Roentgenium [Rn] 5f ¹⁴ 6d ⁹ 7s ²	(285) 5800 1.04 Cn Copernicium [Rn] 5f ¹⁴ 6d ¹⁰ 7s ²	(284) 5800 1.04 Nh Nihonium [Rn] 5f ¹⁴ 6d ¹⁰ 7s ² 7p ¹	(289) 5800 1.04 Fl Flerovium [Rn] 5f ¹⁴ 6d ¹⁰ 7s ² 7p ²	(288) 5800 1.04 Mc Moscovium [Rn] 5f ¹⁴ 6d ¹⁰ 7s ² 7p ³	(292) 5800 1.04 Lv Livermorium [Rn] 5f ¹⁴ 6d ¹⁰ 7s ² 7p ⁴	117 5800 1.04 Ts Tennessine [Rn] 5f ¹⁴ 6d ¹⁰ 7s ² 7p ⁵	(294) 5800 1.04 Og Oganesson [Rn] 5f ¹⁴ 6d ¹⁰ 7s ² 7p ⁶



notes

- 1 kJ/mol = 0.01036 eV.
- all elements are implied to have an oxidation state of zero.

138.9054 5361 1.10 La Lanthanum [Xe] 5d ¹ 6s ²	140.116 5344 1.12 Ce Cerium [Xe] 4f ¹ 5d ¹ 6s ²	140.9076 5278 1.13 Pr Praseodymium [Xe] 4f ³ 6s ²	144.242 5331 1.14 Nd Neodymium [Xe] 4f ⁴ 6s ²	(145) 5400 1.17 Pm Promethium [Xe] 4f ⁵ 6s ²	150.36 5445 1.17 Sm Samarium [Xe] 4f ⁶ 6s ²	151.964 5471 1.17 Eu Europium [Xe] 4f ⁷ 6s ²	157.25 5894 1.20 Gd Gadolinium [Xe] 4f ⁷ 5d ¹ 6s ²	158.9253 5894 1.20 Tb Terbium [Xe] 4f ⁹ 6s ²	162.500 5730 1.22 Dy Dysprosium [Xe] 4f ¹⁰ 6s ²	164.9303 5810 1.23 Ho Holmium [Xe] 4f ¹¹ 6s ²	167.259 5893 1.24 Er Erbium [Xe] 4f ¹² 6s ²	168.9342 5867 1.25 Tm Thulium [Xe] 4f ¹³ 6s ²	173.054 5304 1.03 Yb Ytterbium [Xe] 4f ¹⁴ 6s ²
(227) 4960 1.10 Ac Actinium [Rn] 6d ¹ 7s ²	232.0380 5970 1.30 Th Thorium [Rn] 6d ² 7s ²	231.0358 5989 1.30 Pa Protactinium [Rn] 5f ² 6d ¹ 7s ²	238.0289 5970 1.38 U Uranium [Rn] 5f ³ 6d ¹ 7s ²	(237) 5945 1.36 Np Neptunium [Rn] 5f ⁴ 6d ¹ 7s ²	(244) 5947 1.28 Pu Plutonium [Rn] 5f ⁶ 7s ²	(243) 5780 1.30 Am Americium [Rn] 5f ⁷ 7s ²	(247) 5810 1.30 Cm Curium [Rn] 5f ⁷ 6d ¹ 7s ²	(247) 5810 1.30 Bk Berkelium [Rn] 5f ⁹ 7s ²	(251) 5810 1.30 Cf Californium [Rn] 5f ¹⁰ 7s ²	(252) 5810 1.30 Es Einsteinium [Rn] 5f ¹¹ 7s ²	(252) 5810 1.30 Fm Fermium [Rn] 5f ¹² 7s ²	(258) 5810 1.30 Md Mendelevium [Rn] 5f ¹³ 7s ²	(259) 5810 1.30 No Nobelium [Rn] 5f ¹⁴ 7s ²

Chapter outline

Chapter outline

- Band structure and material classes
- Doping and semiconductor properties
- Semiconductor junctions

4.1 Physics of semiconductors

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7

Chapter outline

- Basic properties of diodes
- Diode modelling
- Diode circuits



4.2 Semiconductor diodes

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41

4. Semiconductor Physics and Diode Characteristics

Chapter outline

- Band structure and material classes
- Doping and semiconductor properties
- Semiconductor junctions

4.1 Physics of semiconductors

Learning objective

- **Explain** how discrete atomic energy levels broaden into bands in solids.
- **Classify** materials as conductors, semiconductors, or insulators by their band gap E_g .
- **Relate** conductivity to charge carrier density and mobility.
- **Explain** n-type and p-type doping and **identify** the majority charge carriers.
- **Describe** the formation of a p–n junction and the depletion zone.
- **Explain** the behaviour of a p–n junction at equilibrium, in forward bias, and in reverse bias

4.1 Physics of semiconductors

Chapter outline

Bohr's atomic model

Band theory

Classification of conductors and band gap

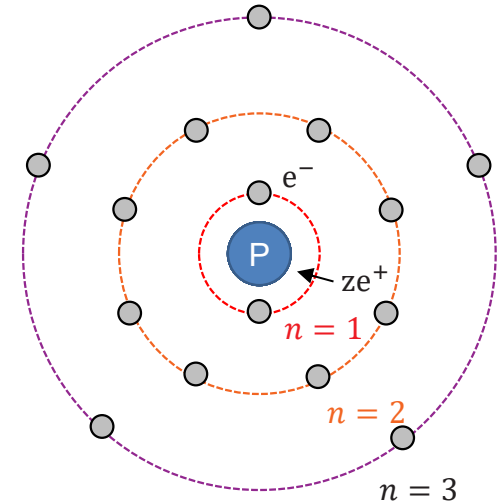
4.1.1 Band structure and material classes

4.1.1 Band structure and material classes

Bohr's Atomic Model

- Extends **Rutherford's** model
 - Electrons occupy **discrete, stationary orbits** around a positively charged nucleus.
 - In these allowed orbits, **electrons do not radiate** energy.
- **Limitations of Bohr's model**
 - Valid mainly for hydrogen-like atoms.
 - Not compatible with modern quantum mechanics.
 - Cannot explain all spectral details.
- Even so, Bohr's model **remains a useful** steppingstone toward understanding electronic states and the band theory of solids.

Tiny positive nucleus, electrons around it, atom mostly empty space.



Bohr's orbital model of the atom.

4.1.1 Band structure and material classes

Electron shell

- Electrons in atoms are described by **four quantum numbers**, satisfying Pauli's principle

Two or more particles cannot occupy the same quantum state!

- Principal quantum number $n = 1, 2, 3, \dots$
- Azimuthal quantum number from $l = 0, 1, \dots, n - 1$
- Magnetic quantum number $m = -l, \dots, l$
- Spin quantum number $s = \pm \frac{1}{2}$ („up“, „down“)

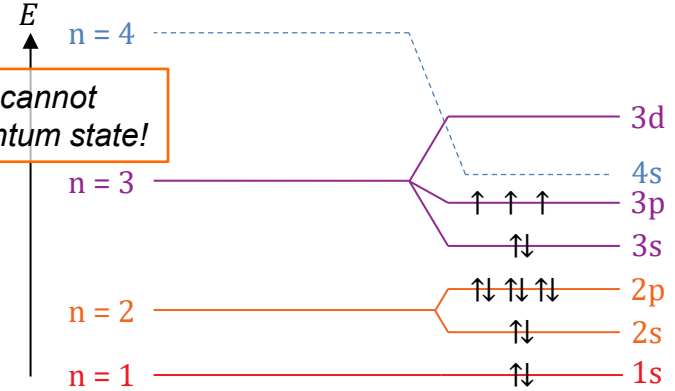
- Available vacancies

- s ($l = 0$): **two** electrons
- p ($l = 1$): **six** electrons
- d ($l = 2$): **ten** electrons
- f ($l = 3$): **14** electrons

- Subshells **filled in ascending energetic order**

- Notation:** Filled orbital + excess: P = [Ne]3s²3p³

- Note:** for $n \geq 4$ the different orbitals start to overlap



		s	p		d				
		l	0	1	2				
		m	0	-1 0 1	-2 -1 0 1 2				
P	n = 3		↑↓	↑ ↑ ↑					
Ne	n = 2		↑↓	↑↓ ↑↓ ↑↓					
He	n = 1		↑↓						

Top: Energetic levels and sublevels of polyelectronic atoms.

Bottom: Electron configuration up to phosphorous.

4.1.1 Band structure and material classes

Exercise: Electron configuration

Write down the electron configuration for Potassium $_{19}\text{K}$ and then for Titanium $_{22}\text{Ti}$. *Note: You may take the periodic table of elements as reference.*



	<i>s</i>				<i>p</i>				<i>d</i>				
<i>l</i>	0				1				2				
<i>m</i>	0	-1	0	1	-2	-1	0	1	2				
<i>n</i> = 4	☐	☐	☐	☐	☐	☐	☐	☐	☐				
<i>n</i> = 3	☐	☐	☐	☐	☐	☐	☐	☐	☐				
<i>n</i> = 2	☐	☐	☐	☐									
<i>n</i> = 1	☐												

4.1.1 Band structure and material classes

Band theory

- Isolated atoms: electrons on **discrete energy levels**.
- Few atoms form **molecules** (covalent bonds, hybridisation) or **ionic crystals** (ionic bonds).
- In a solid with many atoms ($N \approx 10^{22}$) each atomic level (Eigenstate) splits into N levels, closely spaced levels, forming **electronic bands**.
 - ┐ Highest occupied band: **valence band**
 - ┐ Lowest unoccupied band: **conduction band**
 - ┐ **Band gap** in between: distinction between **conductors**, **semiconductors** and **isolators**
 - ┐ **Fermi level** E_F : energy with 50 % occupation probability at $T = 0$ K



The energy scale relates to the inverse kinetic energy required to release an electron from its nucleus into vacuum. One electron volt is the energy an electron gathered from an acceleration voltage of 1 V.

$[E] = \text{eV}$, e. g. $1 \text{ eV} = e \cdot 1 \text{ V} = 1.602\,177 \cdot 10^{-19} \text{ J}$

p
—

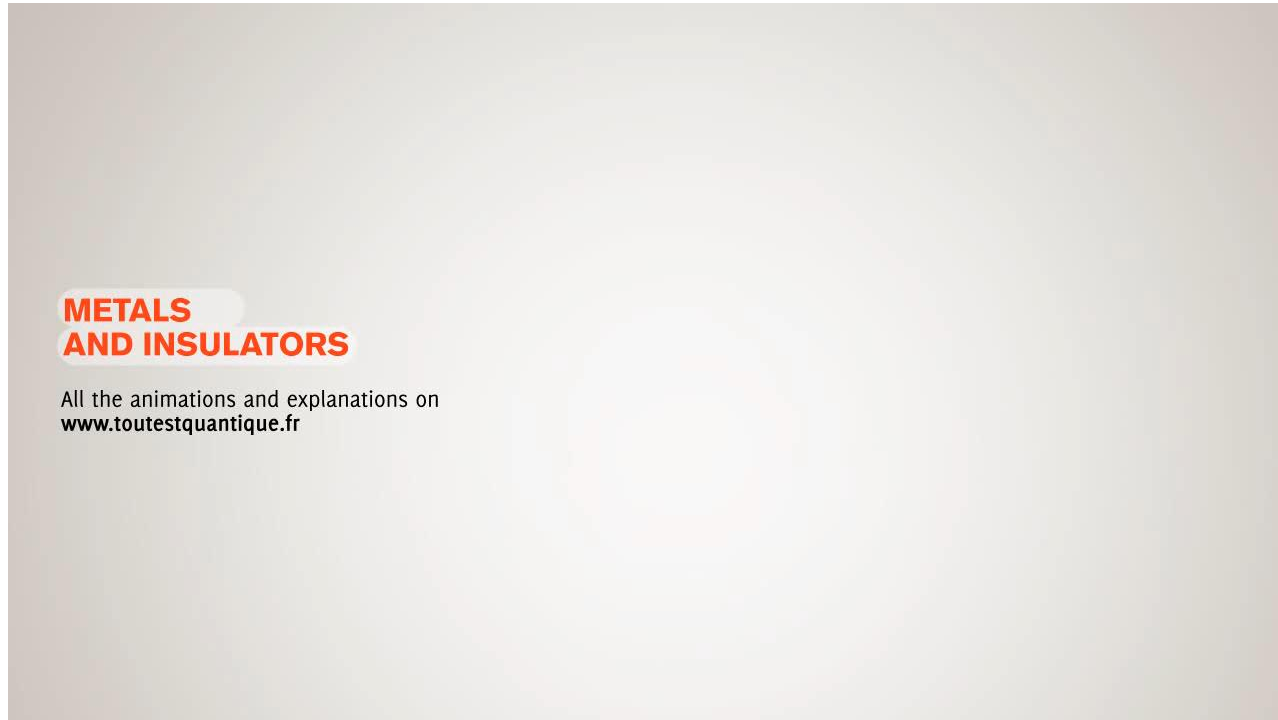
s
—

a Interatomic distance $\longrightarrow$

A hypothetical example of the transition from single atoms to crystal structure.

4.1.1 Band structure and material classes

Band theory – animated



4.1.1 Band structure and material classification

Classification

Conductors ($E_g = 0$ eV)

- Bands partially overlap and only partially filled.
- Electrons can move freely → conductivity

Insulator ($E_g > 4$ eV)

- Valence band is full.
- Conduction band empty.
- Large bandgap → no excitation → no cond.

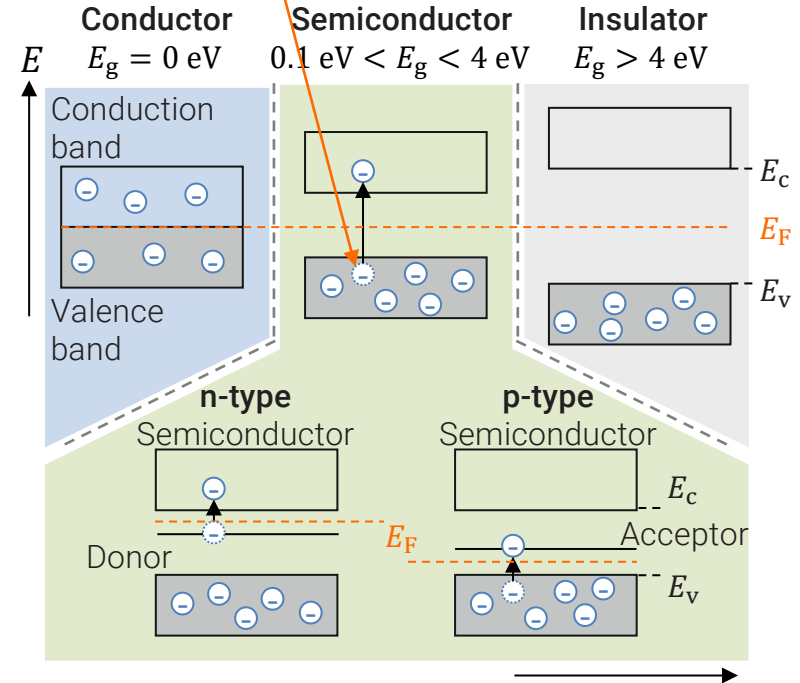
Intrinsic semiconductors (0.1 eV $< E_g < 4$ eV)

- Activation at **moderate temperatures** (25 meV @ 300 K) or via **photon absorption** (2 eV @ 600 nm)
- Mixed conduction (electrons and “holes”)

Extrinsic semiconductors (doped)

- **n-type**: donor, **electron** conduction
- **p-type**: acceptor, **hole** conduction

This is called a defect electron or hole. It behaves like a positive electron.



Classification of materials by their band structure.

4.1.1 Band structure and material classes

Fermi-Dirac distribution

- In a solid, electrons can only **occupy certain allowed energy states**.
- Whether a state with energy E is occupied is described by the **Fermi-Dirac distribution**:

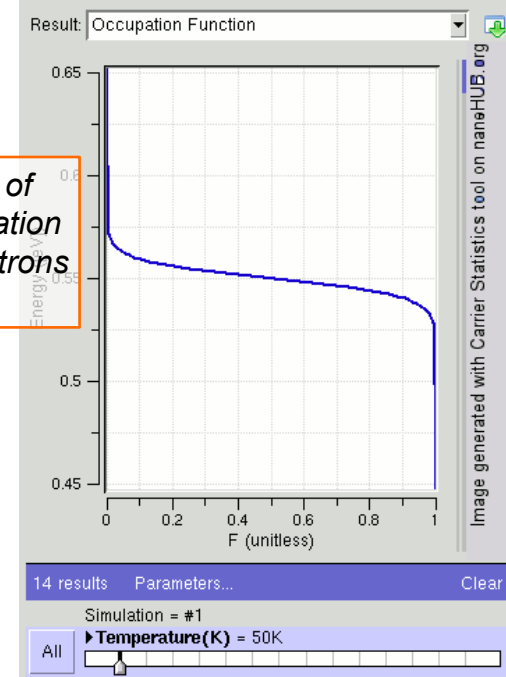
$$f(E) = \frac{1}{\exp\left(\frac{E - E_F}{k_B T}\right) + 1}$$



Silicon at 300 K has a band gap of about 1.12 eV, yet thermal excitation already creates about 10^{10} electrons per cm^3 in the conduction band.

With absolute temperature T , Boltzmann constant k_B and the Fermi level E_F

- $T = 0$ K: all states below E_F are filled, and all states above E_F are empty
- $T > 0$ K: the sharp edge around E_F becomes smeared out, so some electrons can occupy higher-energy states
- These excited electrons, together with the holes left behind in the valence band, increase the electrical conductivity



Fermi-Dirac distribution vs. energy, for $50 \text{ K} \leq T \leq 375 \text{ K}$.

Chapter outline

Density of states

Charge carrier density

For advanced reading

Appendix to 4.1.1: electronic band structure

Appendix to 4.1.1: electronic band structure

Charge carrier density

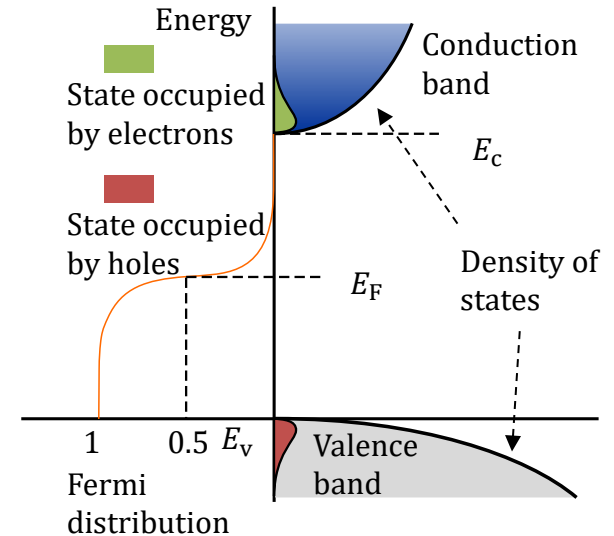
- A fundamental parameter to determine the conductivity
 - ⌈ Metals: $n_0 > 10^{22}$,
 - ⌈ Intrinsic semiconductors: $n_0 \approx 10^6 - 10^{13}$
 - ⌈ Doped semiconductors: $n_0 < 10^{19}$
- In semiconductors the electron density n_0 is described by

$$n_0 \approx T^{\frac{2}{3}} \exp\left(-\frac{E_g}{2 k_B T}\right)$$

With, the band gap energy E_g , Boltzmann constant $k_B \approx 8.617 \cdot 10^{-5} \frac{\text{eV}}{\text{K}}$ and the absolute temperature T in Kelvin

- The hole density p_0 in the valence band is calculated accordingly
- For intrinsic semiconductors in equilibrium the intrinsic carrier concentration is $n_i = n_0 = p_0$

The Boltzmann factor $\exp\left(-\frac{E_g}{2 k_B T}\right)$ defines the probability of an electron having enough thermal energy to jump from the valence band to the conduction band.



Product of a Fermi-Dirac function with a (banded) density of states DOS in an intrinsic semiconductor.

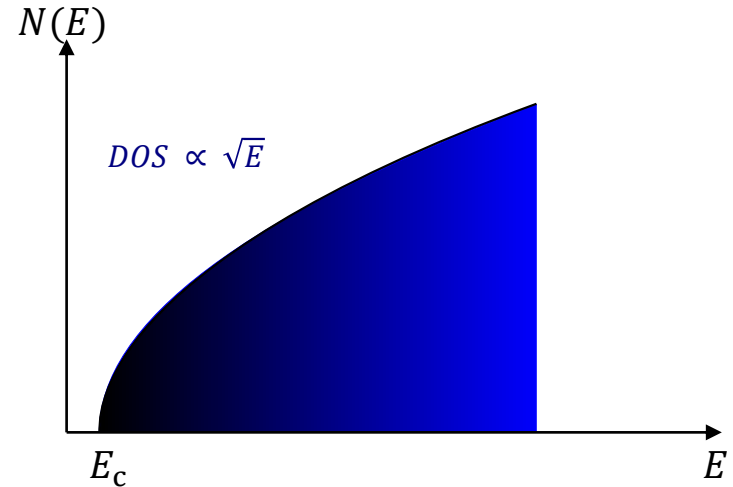
Appendix to 4.1.1: electronic band structure

Density of states

- The **density of states (DOS)** of a system describes the number of modes $N(E)$ per unit frequency range E
- “The DOS describes how many electrons can occupy a certain energy level”
- In a **semiconductor** (low electron density) the electrons are considered to **move freely in a Fermi gas**. The density of states in the conduction band is therefore

$$N(E) = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{E_c - E_F}$$

- 7 With the Planck constant $\hbar = 1.054572 \cdot 10^{-34} \frac{\text{J}}{\text{Hz}}$
- „The number of modes increases with the square root of the energy“
- *Note: The DOS for the valence band is calculated accordingly.*



Free-electron DOS of the conduction band in a semiconductor.

Chapter outline

General characteristics

Doping

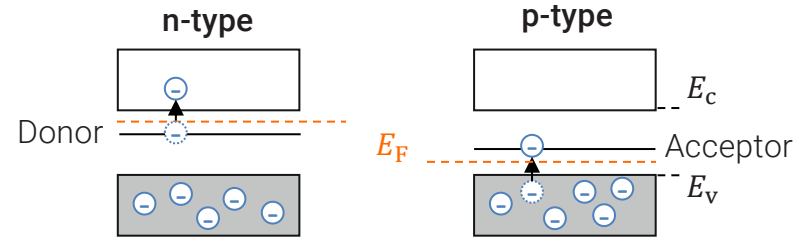
Overview on the most important semiconductor materials

4.1.2 Doping and semiconductor properties

4.1.2 Doping and semiconductor properties

Doping

- **Impurities** in very small fraction to **increase conductivity** by **several orders of magnitudes**
- *Example: n-type doping of Ge (IV)*
 - ⌈ 1 cm^3 of Ge has approx. $4.2 \cdot 10^{22}$ atoms with $2.5 \cdot 10^{13}$ free electrons (and holes).
 - ⌈ 0.001 % arsenic donates 10^{17} free electrons.
 - Conductivity is increase by factor of 10^4 .
- **n-type**: Donor **provides electrons** for the conduction band.
- **p-type**: Acceptor **removes electrons** from the valence band → **hole conduction**.
- Doping **does not change** the position of the valence and conducting **band**.
- Doped semiconductor are **electrically neutral**.
- Electrons and holes are highly diluted (“**electron gas**”) and **chemically diffuse** in the bulk material (analogy: Salt in water).

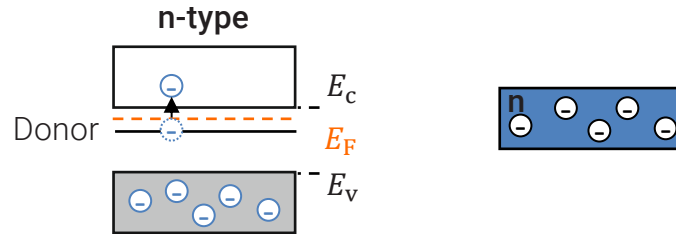


*Left: n-doping with Phosphorous.
Right: p-doping with Aluminium.*

4.1.2 Doping and semiconductor properties

n-type doping

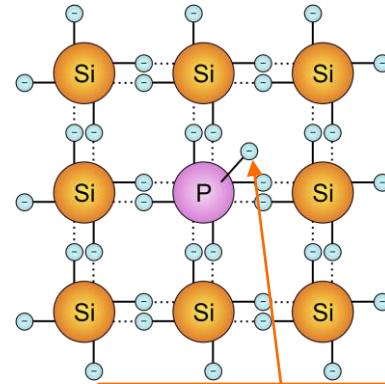
- n-type doping **adds an extra electron** slightly below the conduction band.
- Electrons can be **easily excited into the conduction band**.
- Majority charge carrier: **Electrons**
- n-type dopants for main class IV materials (**Silicon, Germanium**) are from main class V (**Phosphorous, Arsenic, Antimony**)
- The compound stays **electrically neutral!**



Left: Band structure of an n-type doped semiconductor.
Right: Simplified representation of an n-type semiconductor.

Basics of electronics and components

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"This electron was generously donated by Phosphorous! Shall it move around freely!"

— n-type Silicon

Left: Silicon crystal doped with phosphorous.

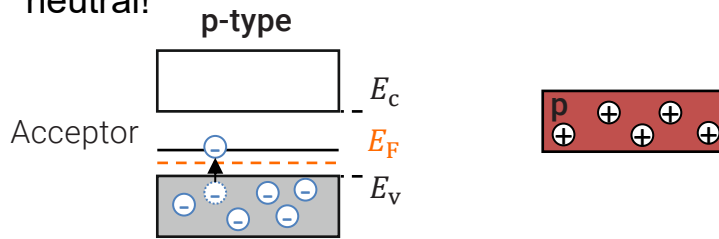
Right: Main groups IV to VI of the periodic table.

III	IV	V
10.811 5 B Boron [He] 2s ² 2p ¹	12.0107 6 C Carbon [He] 2s ² 2p ²	14.0067 7 N Nitrogen [He] 2s ² 2p ³
26.981538 13 Al Aluminium [Ne] 3s ² 3p ¹	28.0855 14 Si Silicon [Ne] 3s ² 3p ²	30.973761998 15 P Phosphorus [Ne] 3s ² 3p ³
69.723 31 Ga Gallium [Ar] 3d ¹⁰ 4s ¹ 4p ²	72.64 32 Ge Germanium [Ar] 3d ¹⁰ 4s ² 4p ²	74.921595 33 As Arsenic [Ar] 3d ¹⁰ 4s ² 4p ³
114.818 49 In Indium [Kr] 4d ¹⁰ 5s ¹ 5p ²	118.710 50 Sn Tin [Kr] 4d ¹⁰ 5s ² 5p ²	121.760 51 Sb Antimony [Kr] 4d ¹⁰ 5s ² 5p ³
204.3833 81 Tl Thallium [Xe] 4f ¹⁴ 5d ¹⁰ 6s ¹ 6p ²	207.2 82 Pb Lead [Xe] 4f ¹⁴ 5d ¹⁰ 6s ² 6p ²	208.9804 83 Bi Bismuth [Xe] 4f ¹⁴ 5d ¹⁰ 6s ² 6p ³
(284) 113 Nh Nihonium	(289) 114 Fl Flerovium	(288) 115 Mc Moscovium

4.1.2 Doping and semiconductor properties

p-type doping

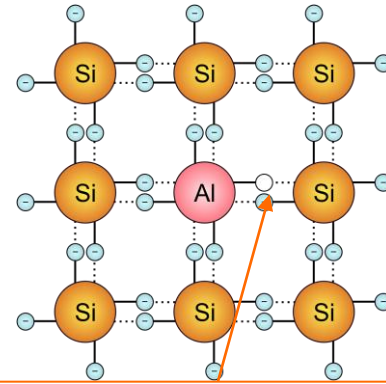
- p-type doping **provides a vacancy for electrons** slightly above the valence band.
- Electrons can be **easily excited from the valence** band.
- Majority charge carrier: **Holes**
- p-type dopants for main class IV materials (**Silicon, Germanium**) are from main class III (**Aluminium, Gallium, Indium**)
- Remember: The compound stays electrically neutral!



Left: Band structure of an p-type doped semiconductor.
Right: Simplified representations of an p-type semiconductor.

Basics of electronics and components

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*“This electron was ripped from nefarious Al-Uminium. The remaining **hole wanders around** in search for the electron.”*

– p-type Silicon

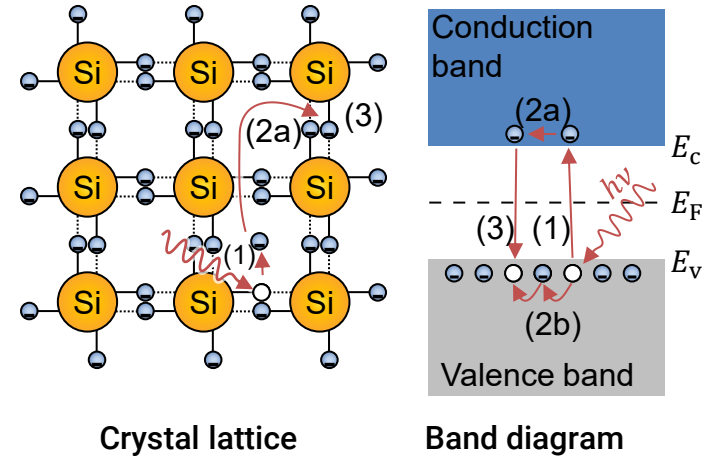
Left: Silicon crystal doped with aluminium.
Right: Main groups IV to VI of the periodic table.

II	IV	V
10.811 5 B Boron [He] 2s ² 2p ¹	12.0107 6 C Carbon [He] 2s ² 2p ²	14.0067 7 N Nitrogen [He] 2s ² 2p ³
26.981538 13 Al Aluminium [Ne] 3s ² 3p ¹	28.0855 14 Si Silicon [Ne] 3s ² 3p ²	30.973762 15 P Phosphorus [Ne] 3s ² 3p ³
69.723 31 Ga Gallium [Ar] 3d ¹⁰ 4s ¹ 4p ²	72.64 32 Ge Germanium [Ar] 3d ¹⁰ 4s ² 4p ²	74.92160 33 As Arsenic [Ar] 3d ¹⁰ 4s ² 4p ³
114.818 49 In Indium [Kr] 4d ¹⁰ 5s ¹ 5p ²	118.710 50 Sn Tin [Kr] 4d ¹⁰ 5s ² 5p ²	121.760 51 Sb Antimony [Kr] 4d ¹⁰ 5s ² 5p ³
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(284) 113 Nh Nihonium	(289) 114 Fl Flerovium	(288) 115 Mc Moscovium

4.1.2 Doping and semiconductor properties

Charge transport in semiconductors

- **Electrons** in conduction band
 - ⌈ Generation and recombination thermally or optically activated
 - ⌈ Low charge density → sensitive to electric and magnetic fields
- **Defect electrons / holes** in valence band
 - ⌈ Paired with excited electron
 - ⌈ “positive charge and negative mass”



- 1) **Electron-hole-pair** generation via photon absorption
 $h\nu \geq E_G$
- 2a) **Electron movement in conduction band**
- 2b) **Hole movement** via position change of electrons in **valence band**
- 3) **Recombination**

*Left: Generation, movement of a electron-hole-pair.
Right: Schematic in the band diagram.*

4.1.2 Doping and semiconductor properties

Materials and compounds

Class	Material	Type	Band gap @ 300 K in eV	Application
Element	C (diamond)	indirect	5.46–5.6	Optical
	Si	indirect	1.12	Solar cells and for integrated circuits
	Ge	indirect	0.67	Substrate for multijunction PV cells
	Se	direct	1.74	Rectifiers
IV-IV	SiC 4H	indirect	3.28	High-voltage and high-temperature applications
III-V	InP	direct	1.27	Substrate for InGaAs, Optoelectronics
	InAs	direct	0.355	Infrared detectors for 1–3.8 μm , quantum dots
	InSb	direct	0.17	Infrared detectors and thermal imaging sensors, 200 GHz transistors
	$\text{In}_x\text{Ga}_{1-x}\text{N}$	direct	0.69–3.37	Blue and green LEDs
	$\text{In}_x\text{Ga}_{1-x}\text{P}$	indirect	2.26	Red LEDs
	GaSb	direct	0.69	Infrared detectors and LEDs
	GaAs	direct	1.42	Substrate for III-V semiconductors, near IR LED, fast electronics, solar cells etc.
	AlGaInP	direct	1.8–2	Red, orange, green, and yellow LEDs and diode lasers
II-VI	BN	direct	5.96	Potentially for ultraviolet LEDs
	TiO ₂ anatase	indirect	3.2	Photocatalytic, n-type
	TiO ₂ rutile	direct	3	Photocatalytic, n-type
	ZnO	direct	3–4	Photocatalytic, tunable bandgap, window coating
	CdTe	direct	1.49	Thin film solar cells, quantum dots

4.1.2 Doping and semiconductor properties

Recap: charge carrier mobility

- The charge carrier **mobility** μ is defined by the drift velocity $\vec{v}_d$ of an **electron** or **hole** in an electric field $\vec{E}$

$$\mu = \frac{\vec{v}_d}{\vec{E}} \quad [\mu] = \frac{\text{m}^2}{\text{V s}}$$
- In solids, mobility depends strongly on temperature because scattering limits carrier motion
 - ⌋ **Low** temperatures: scattering by impurities and **defects** dominates.
 - ⌋ **Higher** temperatures: scattering by **phonons** (lattice vibrations) dominates.
 - ⌋ Carrier-carrier scattering is often less important.
- **Trend:** in many semiconductors, mobility decreases as temperature rises because phonon scattering increases.
- In heavily doped materials, impurity scattering can dominate at low temperature, so the temperature dependence can be more complex.

	Mobility at 300 K in $\frac{\text{cm}^2}{\text{V s}}$	
Material	Electrons	Holes
Metals	≈ 50	-
Si (intrinsic)	1 400	450
Ge	3 900	1900
GaAs	9 200	400
C nanowires	100 000	-
Graphene on SiO2	10 000	-
Graphene on Cu	350 000	-
2D electron gas	350 000	-

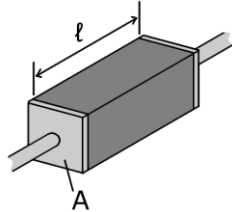
*Electron and hole mobility of **metals**, **semiconductors** and **experimental materials**.*

4.1.2 Doping and semiconductor properties

Recap: electrical conductivity

- The **resistance** R of an element is the product of its inverse conductivity κ and the ratio of length l over area A .

$$R = \frac{1}{\kappa} \frac{l}{A} \quad [\kappa] = \frac{1}{\Omega \text{ m}} = \text{S/m}$$



A piece of resistive material with electrical contacts.

- For **semiconductors** conductivity now also includes electron vacancies with their individual **mobility** μ and **charge carrier density** n and p .

$$\kappa = e (n \mu_e + p \mu_h)$$

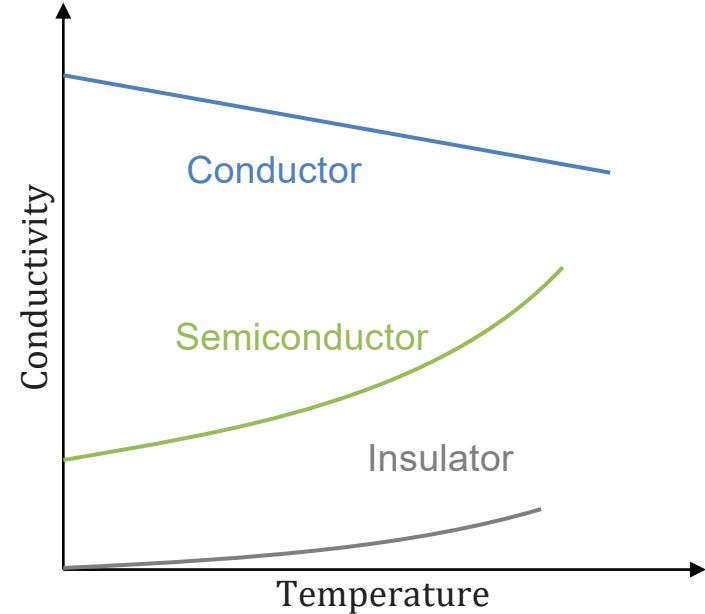
- Note: Find more information about charge carrier density in Appendix to 4.1.1.*

Classification	Material	κ in S/m at 290 K
Metallic conductor $E_g = 0 \text{ eV}$ $n_0 \approx 10^{22}$	Silver Ag	$6.30 \cdot 10^7$
	Iron Fe	$1.03 \cdot 10^7$
	Electrical steel	$2.17 \cdot 10^6$
Non-metallic cond. $E_g \approx 0 \text{ eV}$ $n_0 \approx 10^{24}$	Graphite C (parallel to basal planes)	$> 2 \cdot 10^5$
Semiconductor $0.1 \text{ eV} < E_g < 4 \text{ eV}$ $10^6 < n_0 < 10^{13}$	Germanium Ge	2.17
	Silicon Si (intrinsic)	$4.35 \cdot 10^{-4}$
	GaAs (intrinsic)	10^{-8}
	Silicon Si (doped)	$10^{-4} - 10^4$
	GaAs (doped)	$< 10^3$
Insulator $E_g > 4 \text{ eV}$	Distilled water	$4.2 \cdot 10^{-5}$
	Glass	$10^{-15} - 10^{11}$
	Air	$\approx 10^{-15} - 10^{-9}$
	PTFE (Teflon)	$10^{-25} - 10^{-23}$

4.1.2 Doping and semiconductor properties

Temperature effect

- Mobility and carrier density both depend on temperature.
- In semiconductors, conductivity often increases with temperature mainly because carrier density rises strongly.
- In metals/conductors, conductivity usually decreases with temperature because mobility drop.
- Insulators show a weak temperature dependency.



General temperature-conductivity behaviour of conductors, semiconductors and insulators.

Chapter outline

p-n junction

Metal-semiconductor junction

Zener effect & avalanche effect

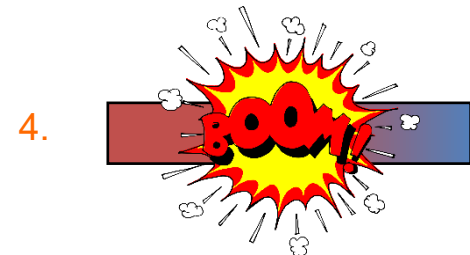
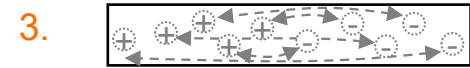
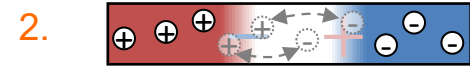
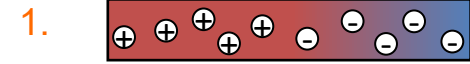
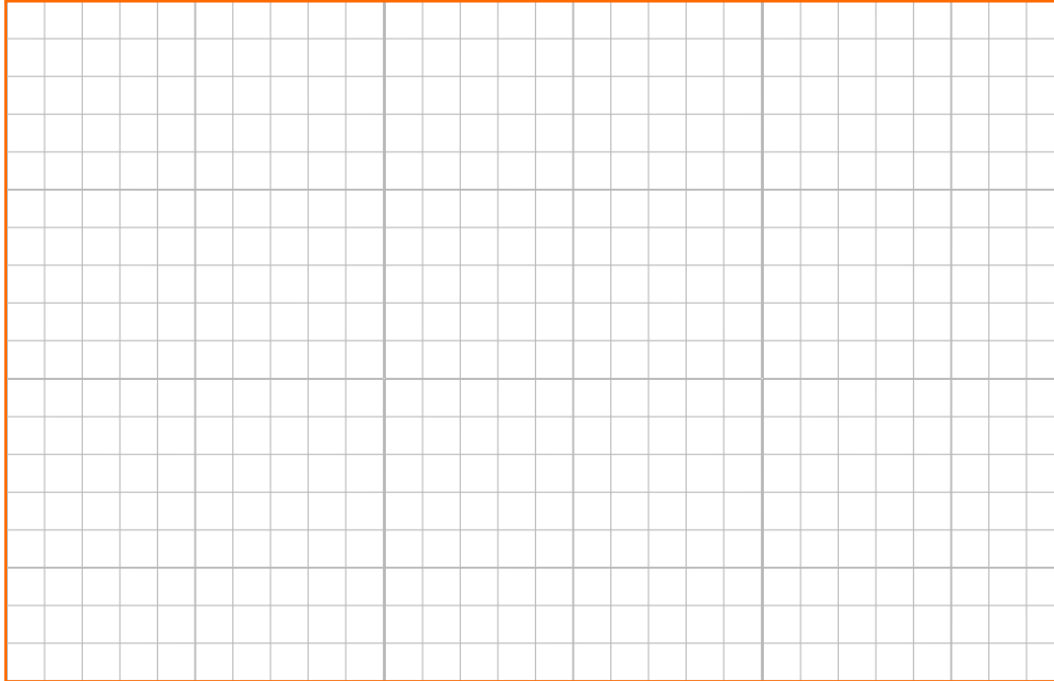
Inner photoelectric effect

4.1.3 Semiconductor junctions

4.1.3 Semiconductor junctions

Peer instruction: Unbiased p-n junction 1/3

What will happen, when you bring a p-type and a n-type semiconductor in contact?

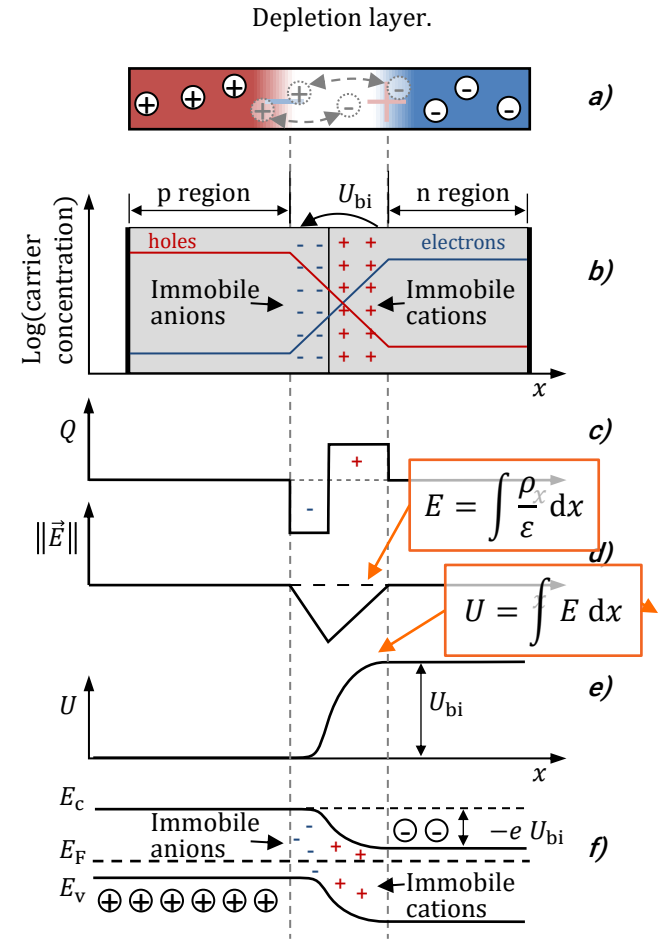


4.1.3 Semiconductor junctions

p-n junction at equilibrium (zero bias)

- **p–n junction:** contact between p-type and n-type semiconductors.
- **Majority carriers** diffuse across the junction and recombine in the opposite region.
- Immobile ionized dopants remain and create an electric field.
- This field forms the **depletion region** and opposes further diffusion.
- The bands bend until the **Fermi level is constant** across the junction.
- At equilibrium, diffusion and drift cancel each other.
- The built-in voltage U_{bi} depends on temperature and doping.

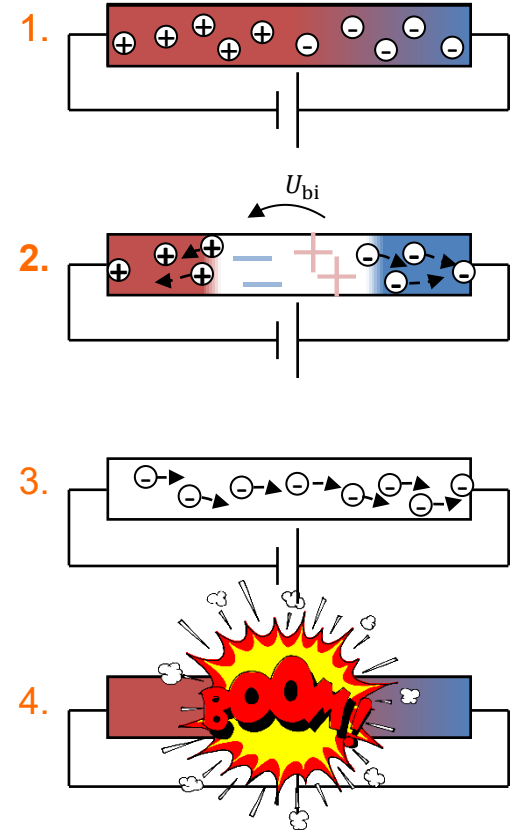
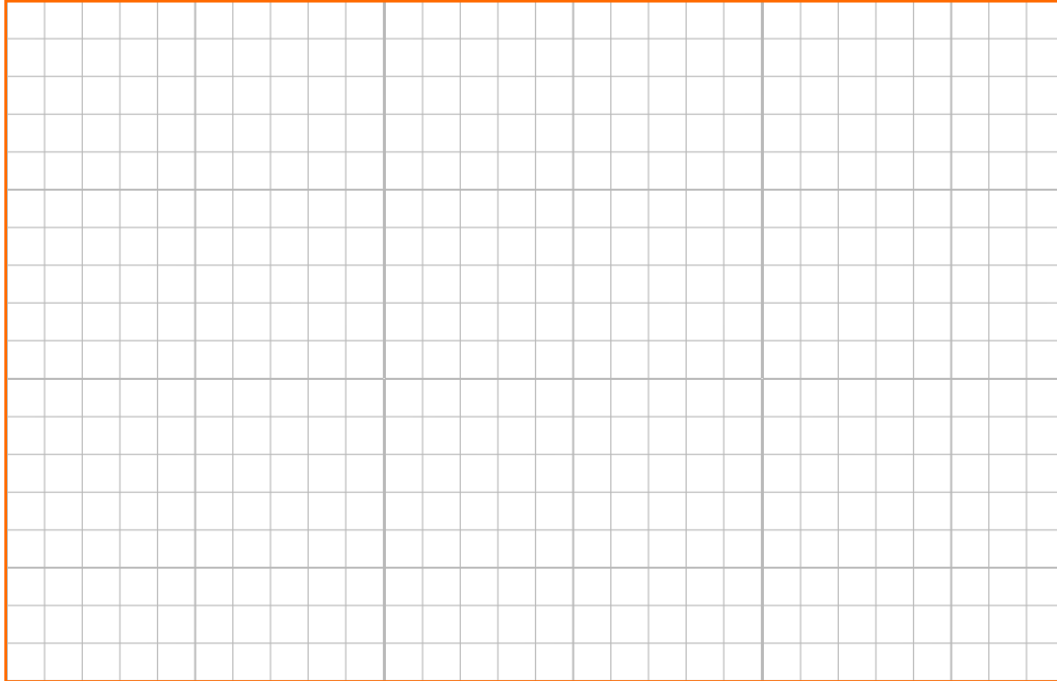
- a) Simplified schematic of a p–n junction at equilibrium.
 b) Detailed schematic.
 c) Space-charge distribution.
 d) Potential variation.
 e) Energy-band diagram.



4.1.3 Semiconductor junctions

Peer instruction: p-n junction in reverse bias 2/3

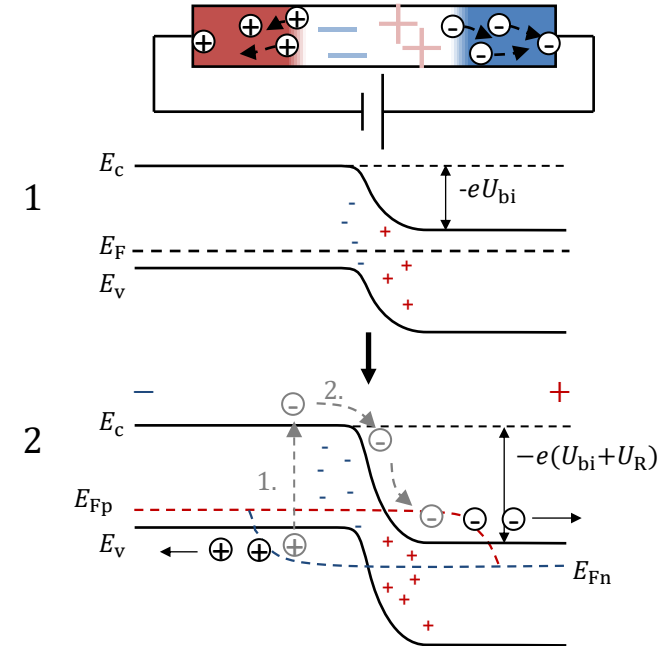
Imagine you connect the p-type with the negative and the n-type with the positive terminal of a battery?



4.1.3 Semiconductor junctions

p-n junction in reverse bias

- A p-n junction with **negative voltage** $U_D < 0$, i. e. the **anode is at lower potential than the cathode**, is in **reverse biased**.
- Charge carriers are **pulled** from depletion zone towards the positive and negative **terminals**.
 - ⌈ **Electric field increases**.
 - ⌈ **Energetic barrier and depletion zone width increases**.
 - ⌈ Diode **capacitance decreases**.
- Electrons **can not run up**, holes **not run down** the slope → the semiconductor is **insulating**.
- When an **electron-hole-pair** is generated **near the depletion zone**, minority charge carrier drift down/up the slope. → **reverse leakage current**.



Schematic of p-n junction in reverse bias (**top**). Transition from equilibrium (**middle**) to reverse bias (**bottom**). Electron-hole-pair generation (1.) and drift (2.) leads to leakage current.

4.1.3 Semiconductor junctions

Breakdown at high reverse bias

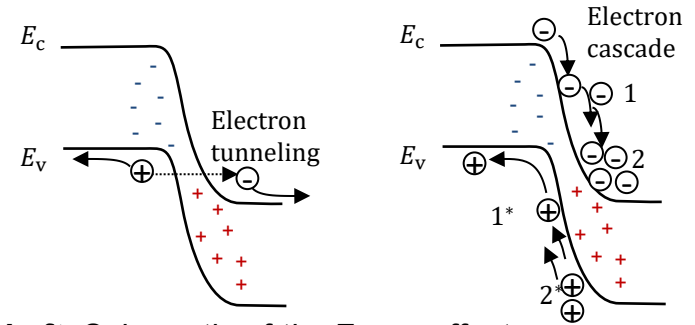
- Under high reverse bias the current increases rapidly → **break down**
- Rapid current increase when operating at **high reverse bias**
- Two competing mechanisms

┐ Zener effect

- Based on: **electron tunnelling** through depletion zone
- Highly-doped p-n junction
- Desired due to the **very well controlled** Zener voltage.

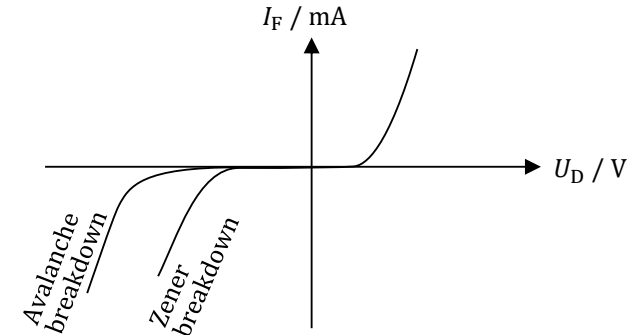
┐ Avalanche effect

- Based on: **cascade effect** of high-momentum electrons
- Normally results in diode destruction.
- Normal mode of operation if designed as **avalanche diode**.



Left: Schematic of the Zener effect.

Right: Schematic of the avalanche effect.

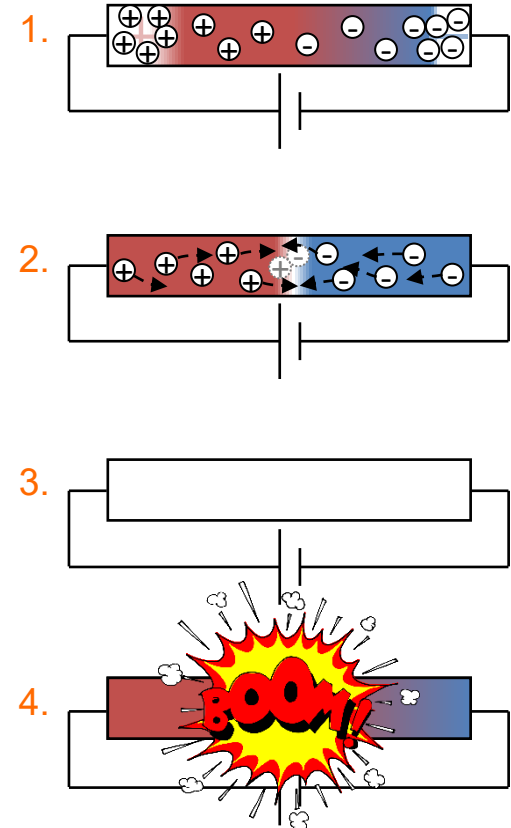
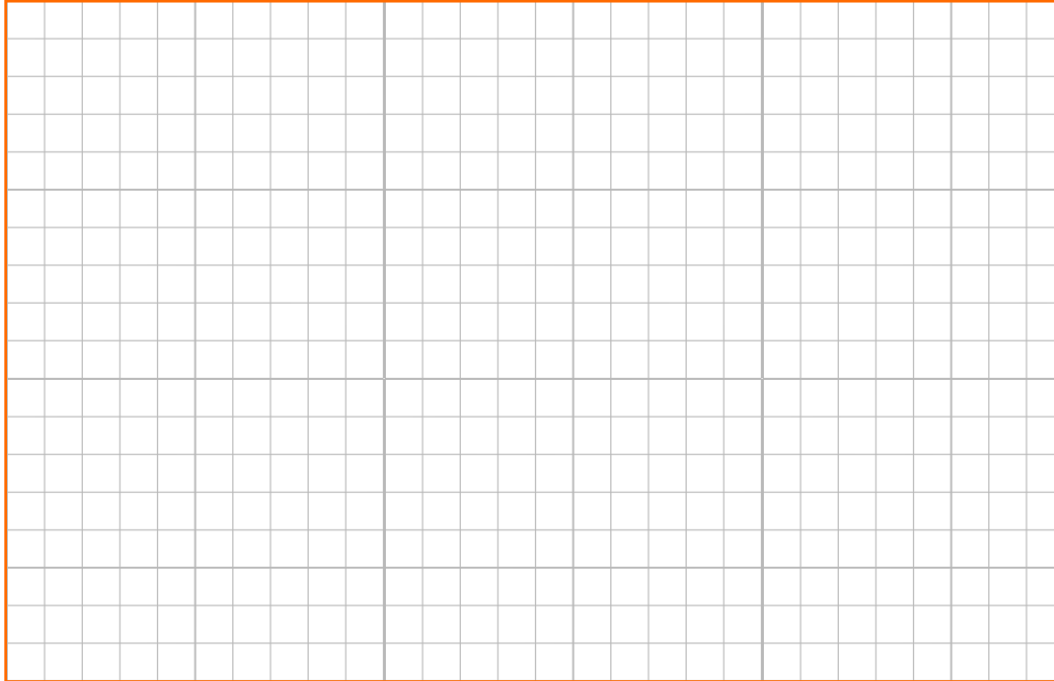


The I-V curve for a diode showing avalanche and Zener breakdown.

4.1.3 Semiconductor junctions

Peer instruction: p-n junction in forward bias 3/3

Now reverse the polarity: p-type is connected to the positive and n-type with the negative terminal of the battery?

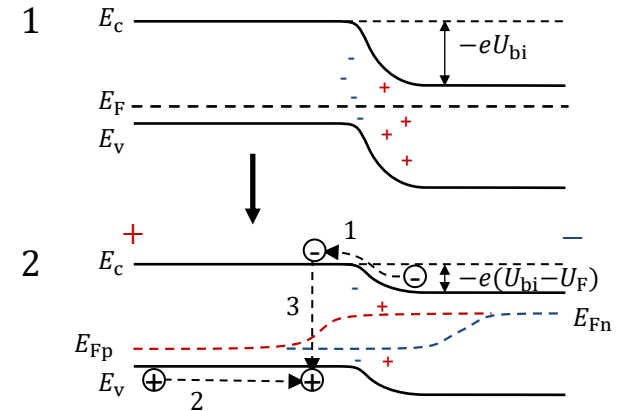
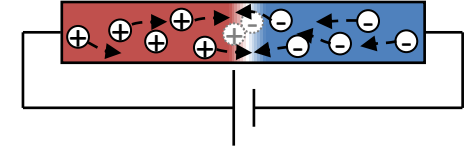


Answer 2 correctly describes the p-n junction in forward bias. Answer 4 may be correct for very high forward voltages!

4.1.3 Semiconductor junctions

p-n junction in forward bias

- A p-n junction with **positive voltage** $U_D > 0$, i. e. the **anode is at higher potential than the cathode**, is under **forward bias**.
- ⌋ **Potential barrier reduced** or is **reversed**.
- ⌋ **Depletion zone width** decreases.
- **Majority** charge carrier can cross the junction
 1. **Electron injection** (diffusion) from n-type into p-type
 2. **Hole** injection from p-type into n-type
- **After crossing the junction, these carriers become minority carriers and recombine with the local majority carriers.**
- Carrier concentration increases exponentially with $U_F \rightarrow$ the semiconductor becomes **highly conductive**.
- *Note: The built-in voltage and the applied forward voltage are related, but they are not identical.*



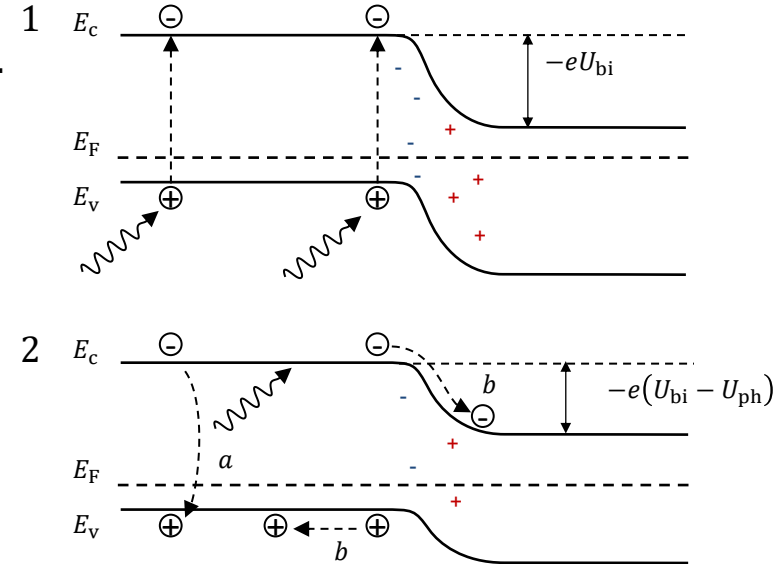
Schematic of p-n junction in forward bias (**top**). Transition from equilibrium (**top**) to forward bias (**bottom**). (1.) Electrons move to p-region, (2.) hole diffusion. (3.) recombination.

4.1.3 Semiconductor junctions

Inner photoelectric effect

- Photon of sufficient energy $\frac{h c}{\lambda} > E_g$ generates an **electron-hole-pair**.
- GaAs: $\lambda = 850 \text{ nm}$
- Si: $\lambda = 1100 \text{ nm}$
- Electron-hole pair recombine after a brief period of time under emission of a photon (light).
- Near the depletion layer an electron-hole pair is **separated** before it can recombine (mean diffusion length).
- Short circuit:**
 - Photo current to electrode.
 - Application: **Light sensor**.
- Open circuit:**
 - Photo voltage U_{ph} in forward direction.
 - Electric field in depletion layer reduced.
 - Application: **Photovoltaics**

Spectrum visible light:
 $400 \text{ nm} \leq \lambda \leq 780 \text{ nm}$



Operating principle of a photodiode (open circuit).

1: Electron-hole pair generation by photon absorption.

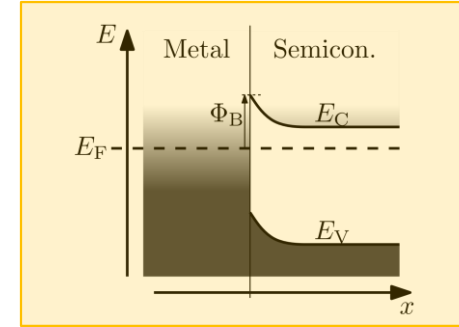
2a Outside depletion-layer: recombination

2b In/near depletion-layer: separation, recombination (not shown) and current to electrodes.

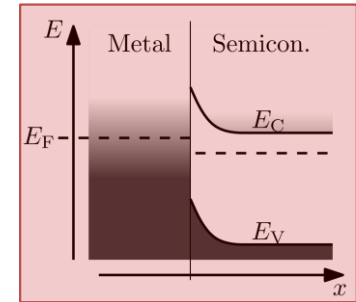
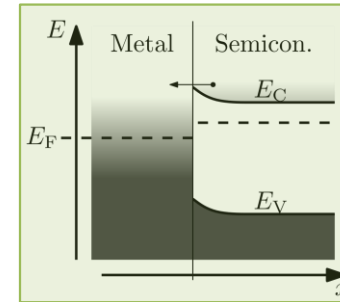
4.1.3 Semiconductor junctions

M-S-junction

- A **metal** in close contact with a **semiconductor** material forms an **energy barrier** Φ_B due to bending of the conduction band of the semiconductor $\rightarrow$ M-S junction
- For $\Phi_B \approx k_B T$: electrons can pass at moderate temperatures $\rightarrow$ non-rectifying **ohmic contact**.
- Characteristics for $\Phi_B \gg k_B T$: rectifying **Schottky barrier**.
 - ⌈ **Zero bias**: no thermal activation, insulator.
 - ⌈ **Forward bias**:
 - ⌊ Thermally excited electrons can pass the barrier.
 - ⌊ Typically, very low threshold voltage ($U_{th} \approx 0.2$ V).
 - ⌊ Current grows exponentially with forward bias.
 - ⌈ **Reverse bias**: Insulating with small leakage current via a few excited electrons.
- *Note: conceptually identical for p-type and n-type Schottky barrier.*



Band diagram for n-type semiconductor Schottky barrier at **zero bias** (equilibrium).



Band diagram for n-type semiconductor Schottky in **forward bias** (left) and **reverse bias** (right).

Summary

- In solids, discrete atomic energy levels broaden into **valence band** and **conduction band**.
- The **band gap** E_g separates both bands and determines whether a material is a conductor, semiconductor, or insulator.
- Materials are classified by E_g : Conductors $E_g \approx 0$, semiconductors $0.1 \text{ eV} < E_g < 4 \text{ eV}$, insulators $E_g > 4 \text{ eV}$
- The conductivity of a semiconductor depends on **charge carrier density** and **mobility**.
- Doping creates **n-type** and **p-type** semiconductors with electrons or holes as majority carriers.
- A p–n junction forms a **depletion zone** and an internal electric field.
- Forward bias reduces the barrier and increases conductivity; reverse bias increases the barrier and suppresses current.

4.1 Physics of semiconductors

Chapter outline

Basic properties of diodes

Diode modelling

Diode circuits

4.2 Semiconductor diodes

Learning objective

- **Explain** the forward and reverse current–voltage characteristic of a diode.
- **Identify** key diode quantities and terminals: anode, cathode, U_{th} , U_{br} , I_S , forward and reverse operation.
- Compare Si, Zener, Schottky and LED in terms of voltage range, speed, leakage, and application.
- **Explain** the meaning of differential resistance r_D .
- **Apply** piecewise-linear diode models and the diode equation to basic circuits.
- **Determine** the operating point graphically using the load-line method.
- **Solve** basic diode circuits with and without load

4.2 Semiconductor diodes

Voltage and current notation of diodes

U_D : signed diode voltage for all diode types. Anode potential minus cathode potential.

U_Z : Zener voltage in reverse breakdown, commonly also diode voltage of Zener diode in reverse operation

I_D : general diode current. Positive in forward direction.

U_F : diode voltage in forward operation, usually $U_F \approx U_D > 0$.

U_R : reverse voltage as a positive magnitude, $U_R = -U_D > 0$.

U_{th} : threshold voltage; rule-of-thumb value for forward conduction.

U_{br} : general breakdown voltage, if you want one symbol beyond Zener diodes.

Default rule: use U_D in equations for general diodes, use U_Z for the reverse operation of Zener diodes

4.2 Semiconductor diodes

Chapter outline

Real and ideal diode current-voltage characteristics

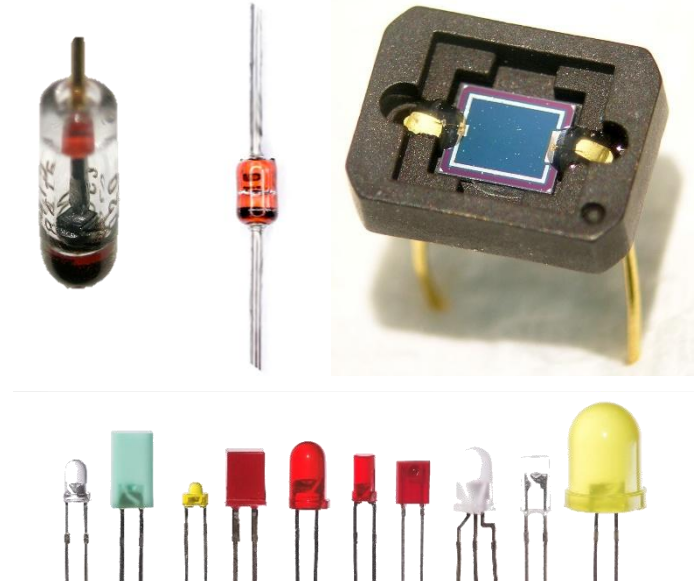
Diode types and applications

4.2.1 Basic properties of diodes

4.2.1 Basic properties of diodes

Characteristics

- Due to their unique properties, **diodes are used in various applications.**
- **Unidirectional, non-linear I-U characteristics**
 - ⌈ AC to DC rectifier
 - ⌈ **Controlled voltage source**
 - ⌈ **Voltage shifter**
 - ⌈ **Base-emitter diode** and **base-collector diode** in bipolar junction transistor
- **Strong temperature dependency** → T-sensor
- Light emission and absorption → light emitting diode **LED** and **photo sensor**



Top left to right: Silicon diode closeup, Zener diode, Photo diode.

Bottom: Various LEDs

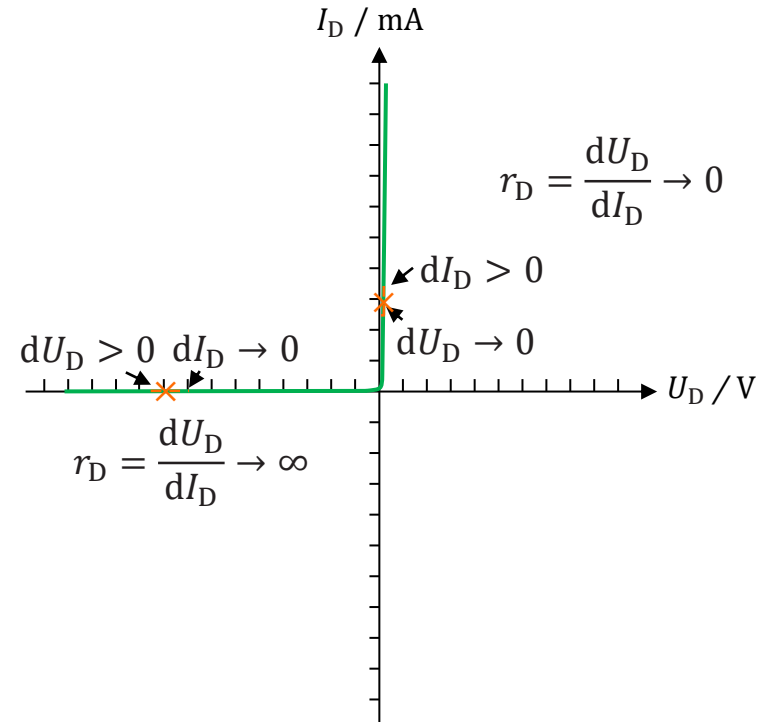
4.2.1 Basic properties of diodes

Current–voltage characteristics

- **Unidirectional, non-linear I-U characteristics around the threshold voltage U_{th}**
 - ⌈ **Reverse bias** ($U_D < 0$): negligible current
 - ⌈ **Forward bias** ($U_D > 0$): exponential current increase
- *Note: noticeable current at around $U_D \approx 0.7\text{ V}$*
- The characteristic is non-linear, the diode has a differential resistance r_D at every **operating point**.

$$r_D = \frac{dU_D}{dI_D}$$

- ⌈ Reverse voltage: $r_D \rightarrow \infty$
- ⌈ Forward voltage: $r_D \rightarrow 0$
- Also called **small-signal resistance**

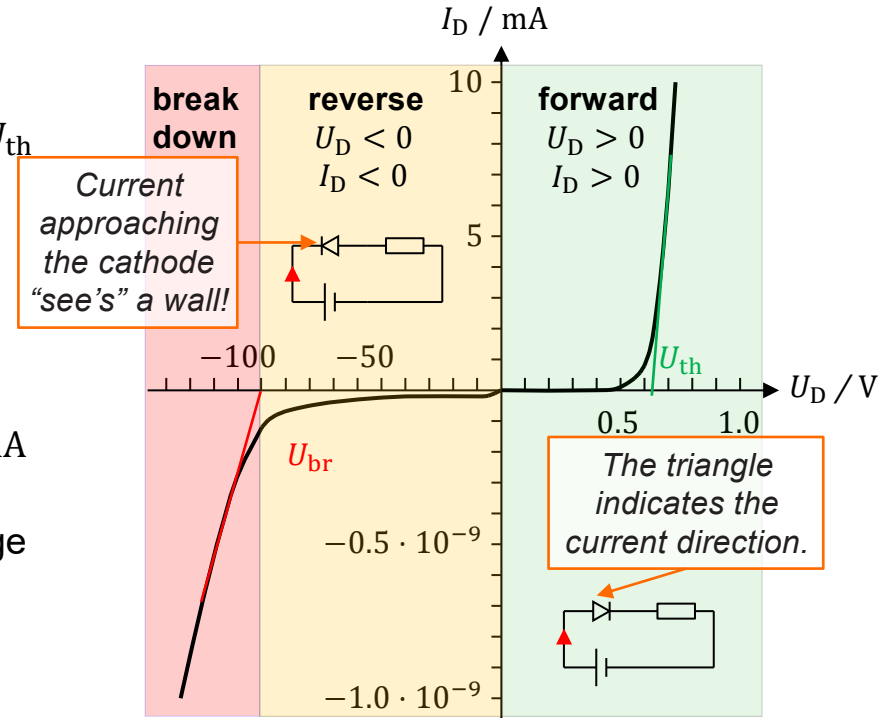


Idealized characteristic curve of a diode.

4.2.1 Basic properties of diodes

Real diode modes of operation

- **Forward mode** $U_D > 0$
 - ⌈ **Positive terminal** is connected to **p-type** (anode).
 - ⌈ Noticeable current around threshold voltage $U_D \approx U_{th}$
 - ⌈ Almost **constant voltage** equal to the threshold voltage and independently of current.
 - ⌈ Very small resistance beyond threshold voltage.
- **Reverse mode** $U_D < 0$.
 - ⌈ **Positive** terminal is connected to **n-type** (cathode).
 - ⌈ Small **leakage current** (reverse saturation, $I_S < 1$ nA for Si-diodes)
 - ⌈ Strong current increase when the breakdown voltage U_{br} is reached.
- *Note: Exact values for the threshold and breakdown voltages do not exist; they are obtained from models, extrapolation, or rules of thumb (for silicon diodes typically ($U_{th,Si} \approx 0.7$ V).*



Characteristic curve of a 1N914 rectifier diode at room temperature.

4.2.1 Basic properties of diodes

Diode types

— Silicon p-n

- Standard diode based on p-n homo- or heterojunction.

- Versatile “**Jack-of-all-trades**” device

— Z-Diode / Zener

- Heavily doped** Si diode using Zener and avalanche breakdown.

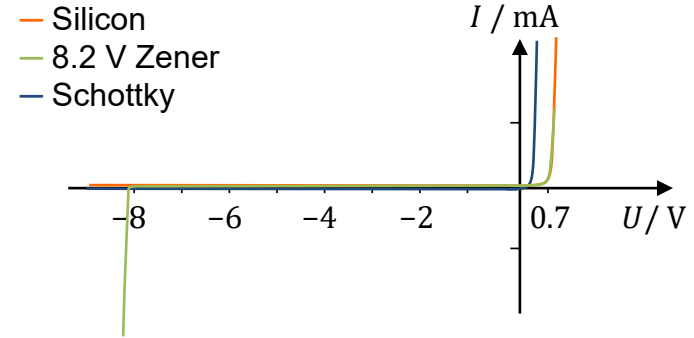
- Forward: normal Si diode ($U_D \approx 0.7 \text{ V}$).

- Reverse: $U_Z = -U_D$, constant ‘Zener voltage’.

- Available in fine voltage increments. e. g. $U_Z = 3.0 \text{ V}, 3.3 \text{ V}, 3.6 \text{ V}, 3.9 \text{ V} \dots$

— Schottky

- Metal–semiconductor junction diode.
- Ultra-fast switching** due to low capacitance.
- Used in high-frequency and, with SiC, high-power circuits.
- Relatively large leakage current in reverse operation.



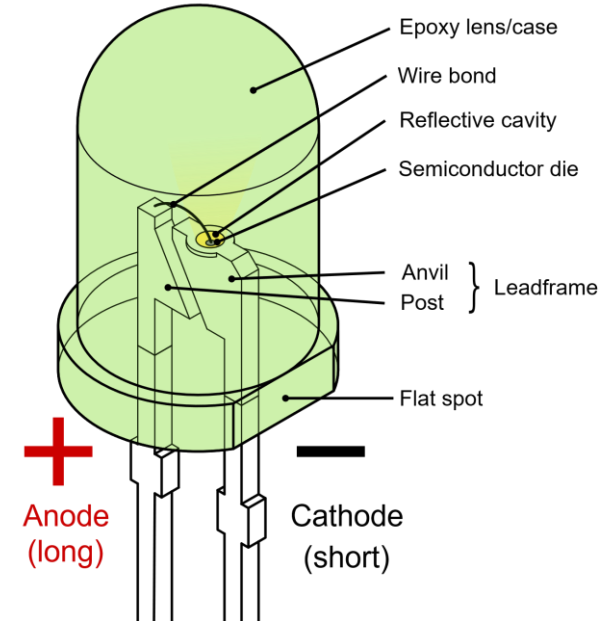
Characteristic curves of Silicon, Zener and Schottky diodes.

Characteristics	p-n	Zener	Schottky	
Type	Si	heavily doped Si	Mo, Pt, etc.	SiC
Threshold voltage U_{th} / V	0.7		0.15 – 0.46	1.4 – 1.8
Breakdown voltage U_{br} / V	70 – 100	2.7 – 200	50	< 1700
Switching time / s	10^{-6}	10^{-9}	10^{-12}	10^{-9}
Leakage current I_S / A	$10^{-12} - 10^{-9}$		10^{-6}	

4.2.1 Basic properties of diodes

Light-emitting diode LED

- P-n diode operated in **forward direction**.
- **Emits light** due to **recombination** of hole-electron pairs.
- **Colour of light corresponds to band gap** $E_g = h f = h c / \lambda$.
- Various colours by different **material-combinations**.
 - ⌈ **Red R: AlGaInP**, **Green G: InGaN**, **Blue B: GaN** and InGaN
 - ⌈ **White : RGB multi-semiconductors** or a layer of light-emitting **phosphor**
 - ⌈ Infrared: GaAs
 - ⌈ UV and near UV: AlGaIn and AlGaInN
- Applications
 - ⌈ General lightning and **illumination**.
 - ⌈ Data **communication**.
 - ⌈ Narrow band **light sensors** in reverse mode.
 - ⌈ Or, when **different threshold voltages** are required.



Parts of a conventional LED.

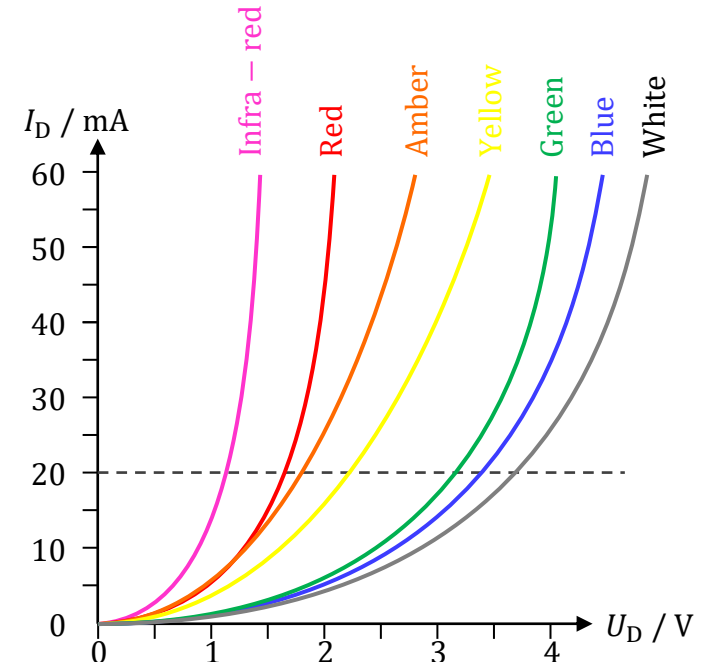
4.2.1 Basic properties of diodes

Light-emitting diode LED

Semiconductor Material	Wavelength / nm	Colour	Threshold voltage / V
GaAs	850-940	Infra-Red	1.2
GaAsP	630-660	Red	1.8
GaAsP	605-620	Amber	2.0
GaAsP:N	585-595	Yellow	2.2
AlGaP	550-570	Green	3.5
SiC	430-505	Blue	3.6
GaN	450	White	4.0



Because different LED colors have very different forward threshold voltages, they can also be used in non-lighting applications, for example as simple voltage references or light sensors.



Light Emitting Diode (LED) Schematic symbol and I-V Characteristics Curves showing the different colours available.

Recap

- Diodes are **non-linear** two-terminal components with strongly direction-dependent behaviour.
- Forward bias: steep current increase; reverse bias: small leakage current until breakdown.
- U_{th} and U_{br} are model parameters, not exact switching points.
Differential resistance r_D depends on the operating point, so a diode is not an ohmic resistor.
- Si, Zener, Schottky, LED, and photodiodes differ mainly in voltage range, switching speed, leakage current, and application.
- Typical applications: rectification, voltage shifting, voltage regulation, light emission, light sensing.

4.2.1 Basic properties of diodes

Chapter outline

Model 1: ideal switch

Model 2: switch + voltage source

Model 3: switch + voltage source + r_D

Diode equation and Load-line method

4.2.2 Diode modelling

4.2.2 Diode modelling

Extrapolating around the operating point

- **Graphical method** to determine **threshold voltage** U_{th} and **differential resistance** r_D around an operating point $Q(U_D, I_D)$.

- **Threshold voltage** U_{th} :

- ⌈ Obtained by linearizing the steep forward region around a chosen operating point and extrapolating to $I_D = 0$.
- ⌈ U_{th} is a **model parameter**, not a unique physical constant.

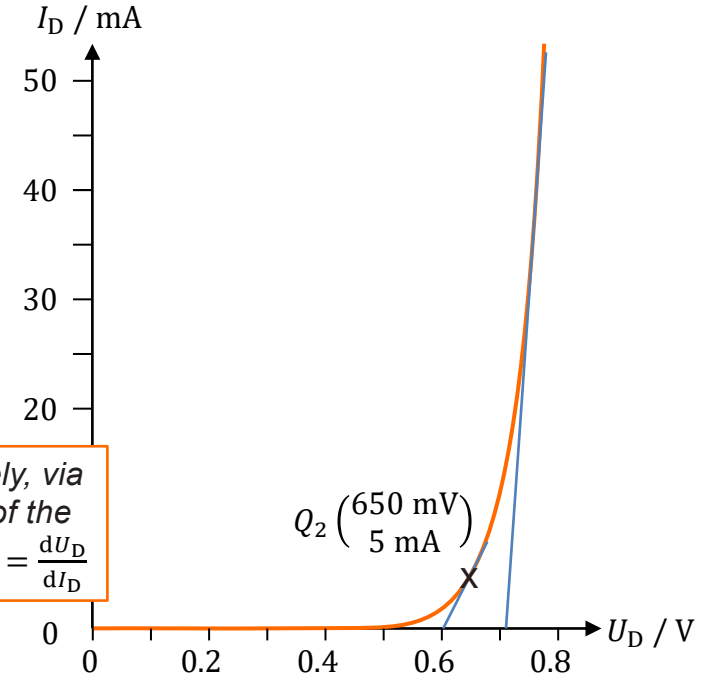
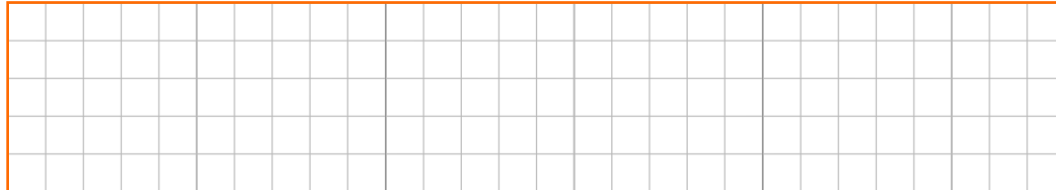
- **Differential resistance** r_D

- ⌈ Symmetrically extrapolate around the Q ; get ΔU_D and ΔI_D .
- ⌈ Get the differential resistance via (linearization around Q)

$$r_D = \frac{\Delta U_D}{\Delta I_D}$$

Alternatively, via the slope of the tangent $r_D = \frac{dU_D}{dI_D}$

- **Exercise:** Determine U_{th} and r_D at $Q_2(650 \text{ mV}, 5 \text{ mA})$.



Current-voltage curves: Characteristic curve of a 1N4001 diode. Linear extrapolation.

4.2.2 Diode modelling

Modelling: Piecewise linear model 1/3

1. Diode modelled as a switch

— **Reverse** bias or “switch open” $U_i < 0$

└ $r_D = \infty \rightarrow$ open circuit

└ $U_D = U_i < 0$

└ $I_D = 0$

— **Forward** bias or “switch closed” $U_i \geq 0$

└ $r_D = 0 \rightarrow$ short circuit

└ $U_D = 0$

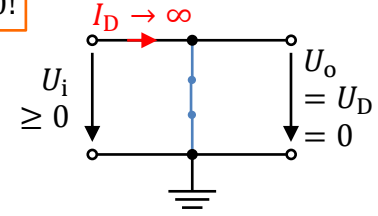
└ $I_D \rightarrow \infty$

— **Use this model if:**

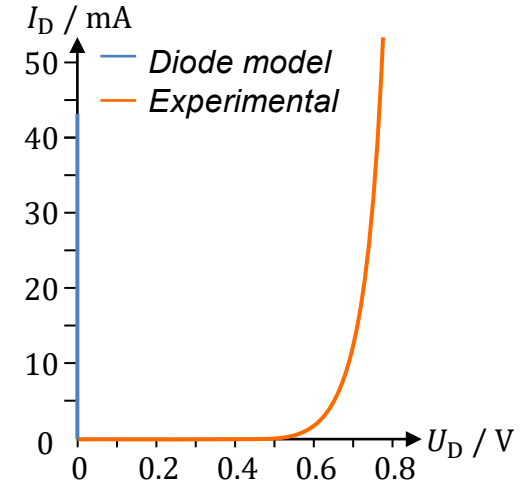
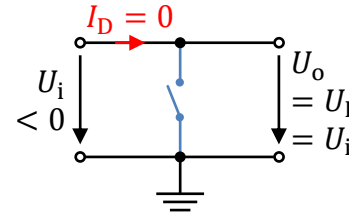
└ Interested in the rectifying properties of diodes.

└ The input voltage is large compared to the diode voltage.

For ideal diodes $U_i = 0$!



Shunt two-port.



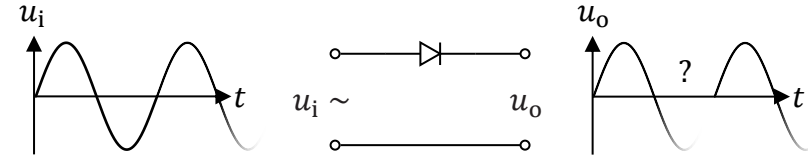
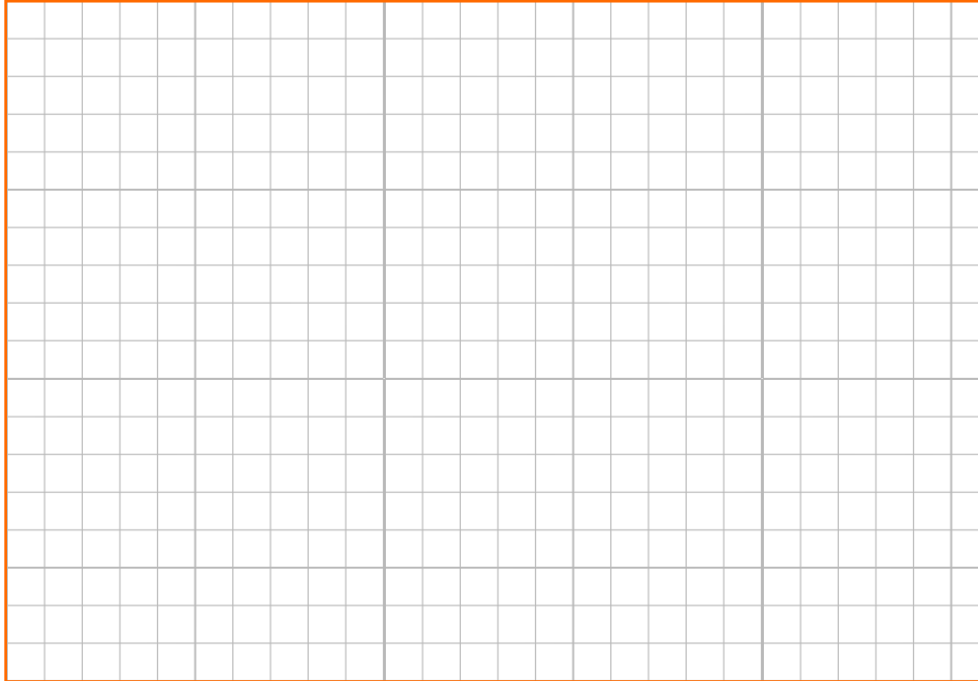
Current-voltage curves: *Characteristic curve of a Si-diode.*
Diode modelled as switch in series and shunt configuration.

4.2.2 Diode modelling

Exercise: Half-wave rectification 1/2

EXAM	Typical Exam Question
•	
•	
•	

Describe effective output voltage U_o of a **series two-port** comprising a single diode in series with an AC input voltage $u_i = 325 \text{ V} \cdot \sin \omega t$. What is the effective output voltage?

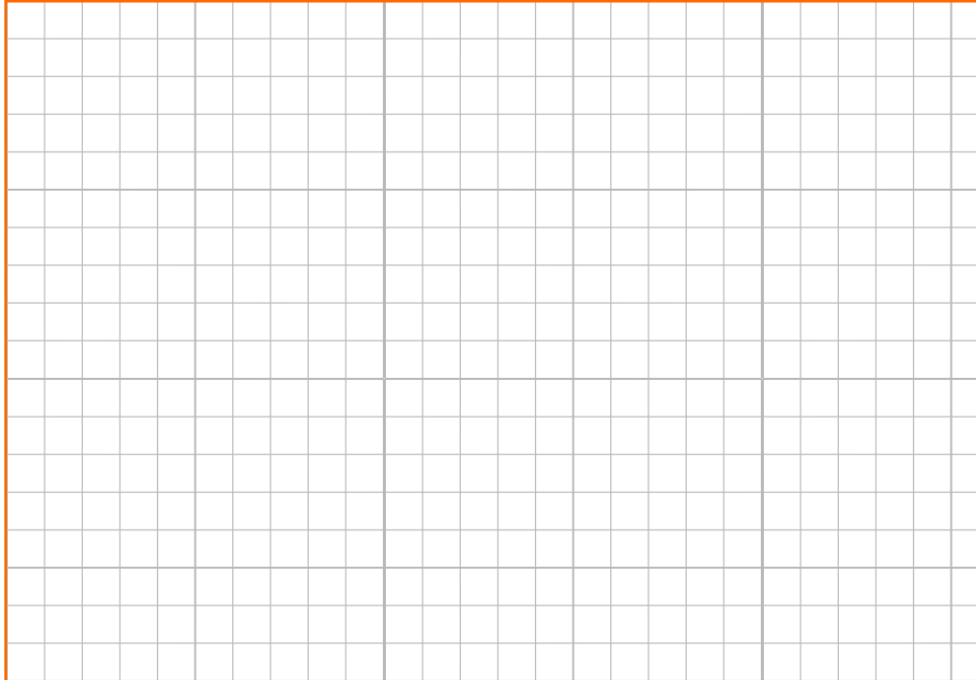


4.2.2 Diode modelling

Example: Half-wave rectification 2/2

EXAM	Typical Exam Question
• —	
• —	
• —	

Now graphically evaluate the output voltage as a function of the input voltage. First construct the output to input characteristics via the so-called transfer curve. Then mirror a sinusoidal wave at this transfer curve.



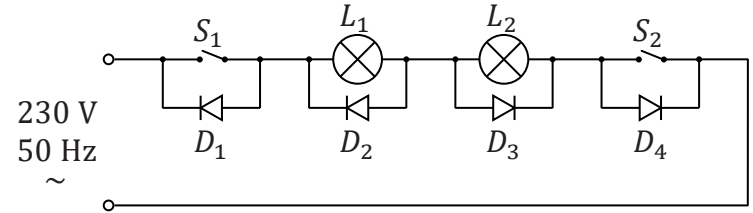
⋮

4.2.2 Diode modelling

Exercise: Diode as a switch

EXAM	Typical
• —	Exam
• —	Question
• —	
• —	

Please describe the state of the lights L_1 and L_2 for the following circuit for all states of the switches S_1 and S_2



4.2.2 Diode modelling

Modelling: Piecewise linear model 2/3

2. Switch + voltage source

- Add an **ideal voltage source** to shift the voltage by the diode threshold voltage U_{th} (offset).
- In reverse mode the diode is open. The diode voltage is equal the input voltage.

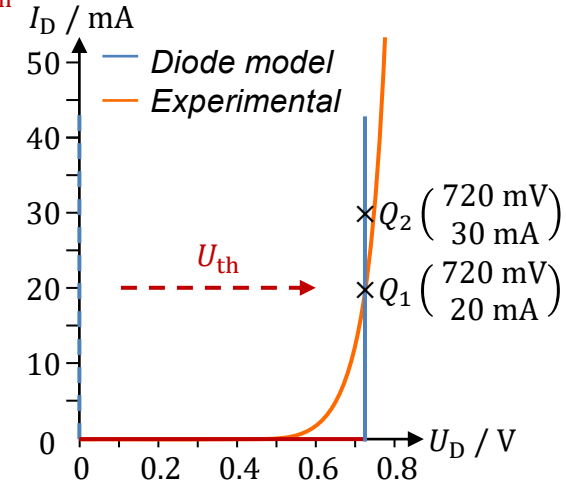
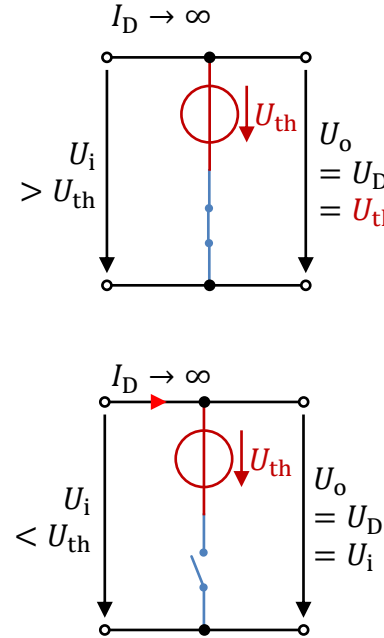
$$U_i < U_{th} \quad \therefore U_D = U_i \quad \therefore r_D \rightarrow \infty$$

- In forward mode the diode voltage is equal to the threshold voltage.

$$U_i > U_{th} \quad \therefore U_D = U_{th} \quad \therefore r_D \rightarrow 0$$

- **Use this model if:**

- 7 Operating with **small-signals**.
- 7 No additional information is provided (see next example) or r_D is given as **zero**.
- 7 You are interested in a **quick-and-dirty circuit design**, e. g. when biasing a transistor.



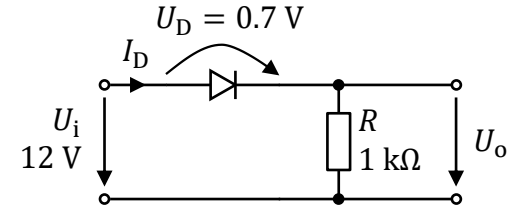
Current-voltage curves: *Characteristic curve of a Si-diode.*
Diode modelled as switch + voltage source.

4.2.2 Diode modelling

Example: Diode as a constant voltage source

EXAM	Typical
•	Exam
•	Question
•	
•	

You are given the following circuit comprising a silicon diode and a resistor ($R = 1 \text{ k}\Omega$) and a DC input voltage $U_i = 12 \text{ V}$. Please calculate the output voltage U_o and the diode current I_D .



4.2.2 Diode modelling

Modelling: Piecewise linear model 3/3

3. Switch + voltage source + resistor

- Add a **resistor** r_D to account for the current-dependent voltage change of the diode near an **operating point** $Q(U_Q, I_Q)$.
- The differential resistance r_D depends on the selected operating point:

$$r_D \approx \frac{\Delta U_D}{\Delta I_D} = \frac{U_D - U_Q}{I_D - I_Q}$$

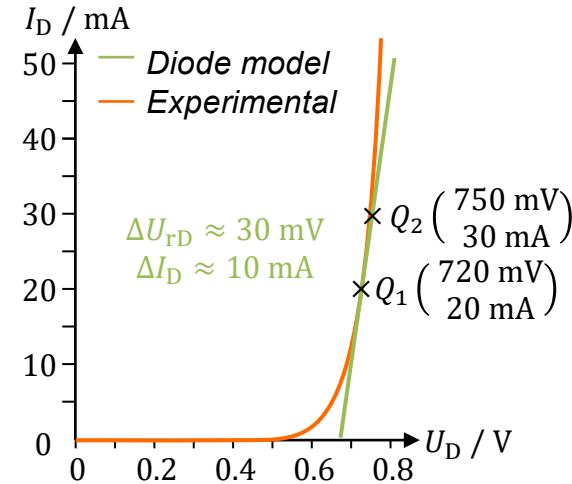
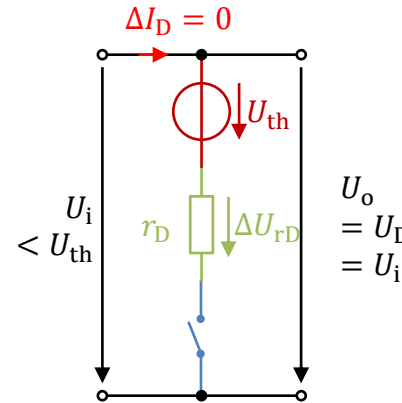
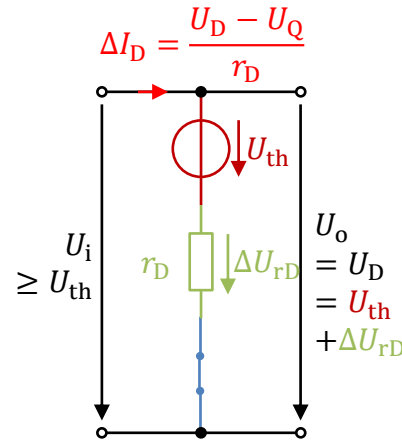
- Linearized model around $Q(U_Q, I_Q)$ via KVL:

$$U_D = U_Q + \Delta U_{rD} = U_Q + r_D \Delta I_D = U_Q + r_D (I_D - I_Q)$$

- Choosing $Q(U_{th}, 0)$ gives the simple form:

$$U_D \approx U_{th} + r_D I_D$$

- **Use this model** when the voltage across the diode resistance can not be neglected.

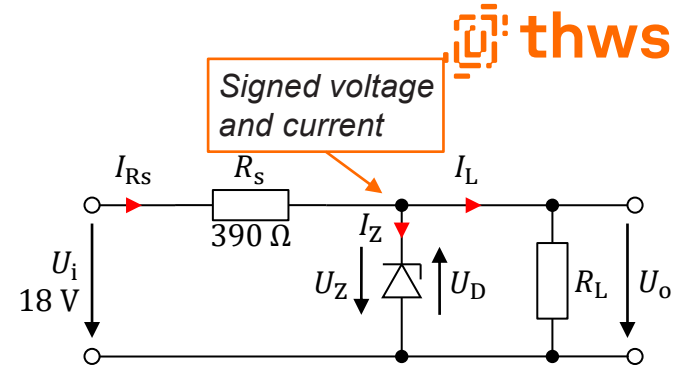


Current-voltage curves: *Characteristic curve of a Si-diode.*
Diode modelled as switch + voltage source.

4.2.2 Diode modelling

Example: Diode modelled as voltage source and resistor

Calculate the current I_{R_S} in the stabilizer circuit for a Zener current $I_Z = 20 \text{ mA}$. The Z-diode is parametrized by the resistance $r_D = 10 \Omega$ at the operating point $Q(6.1 \text{ V}, 5 \text{ mA})$. Compare with the result when neglecting r_D .



4.2.2 Diode modelling

Modelling: Diode equation

- Empirical model developed one of the co-founders of the transistor.
- For Si type and at room temperature.

Boltzman
constant

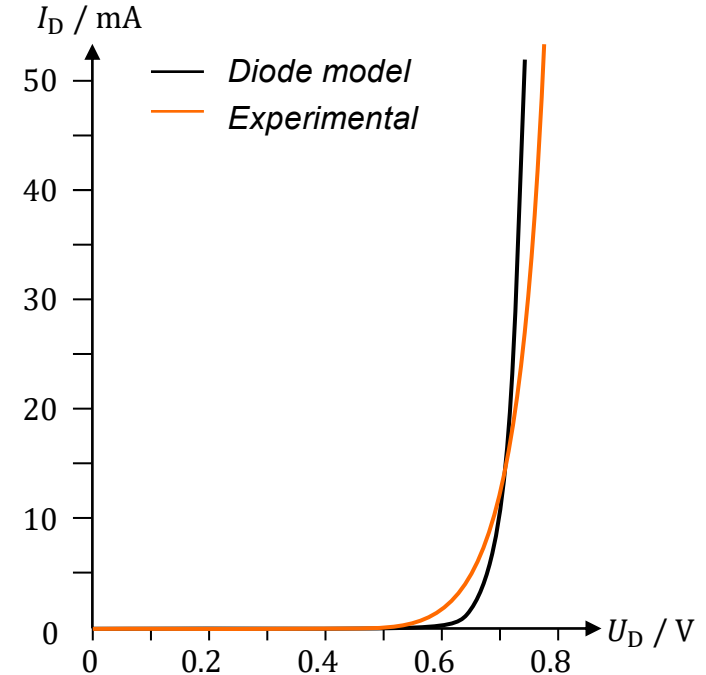
$$I_D = I_S \exp\left(\frac{U_D}{U_T}\right) \quad (I)$$

Energy of electrons
at room temperature.

Where, $U_T = \frac{k_B T}{e} \approx 25.7 \text{ mV}$ is the thermal voltage at room temperature ($T = 298 \text{ K}$).

Elementary charge

- Typical reverse-bias saturation or leakage currents:
 - ⌈ $I_S = 10^{-12} \text{ A} - 10^{-13} \text{ A}$ for Si diodes
 - ⌈ $I_S = 10^{-6} \text{ A}$ for Schottky diodes
- Does not consider internal resistance → works o. k. for moderate forward bias voltage.
- *Note: This model considers that the energy of charge carriers follows the Boltzmann distribution. The probability of excited electrons grows exponentially with the temperature.*



Diode U-I curves: Characteristic curve of a 1N4001 Si-diode and diode parametrized with $U_T = 25.7 \text{ mV}$, $I_S = 10^{-13} \text{ A}$.

4.2.2 Diode modelling

Modelling: Diode equation

- Solving (I) for U_D we get for Si-diodes:

$$U_D = U_T \ln \left(\frac{I_D}{I_S} \right) \quad (II)$$

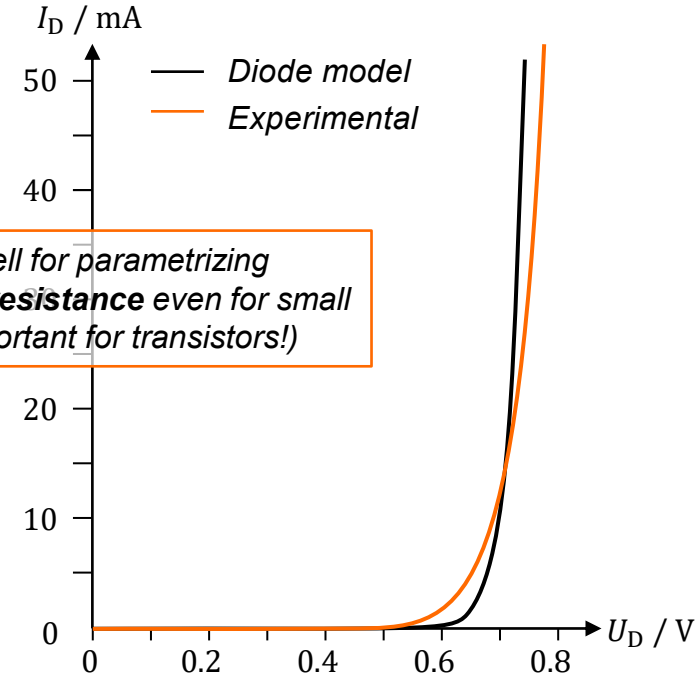
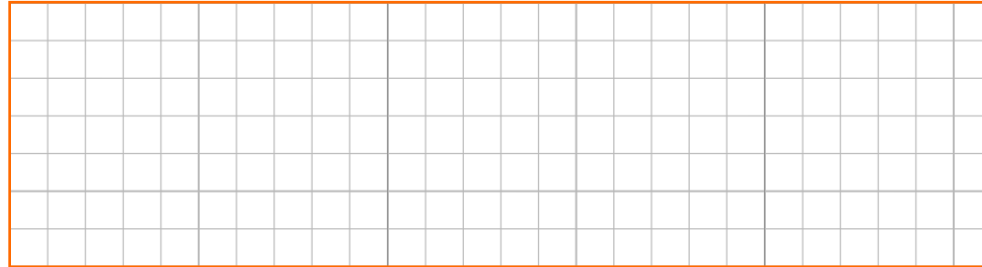
I_S is often off about some 100 mV for small currents.

- Deriving U_D by I_D provides the **small-signal model** of the input resistance around an operating point:

$$r_D = \frac{dU_D}{dI_D} = \frac{U_T}{I_D} \quad (III)$$

*Work very well for parametrizing **differential resistance** even for small signals. (Important for transistors!)*

- Note: We'll need this later for the input resistance of transistors!
- Exercise: Derive equation (III).

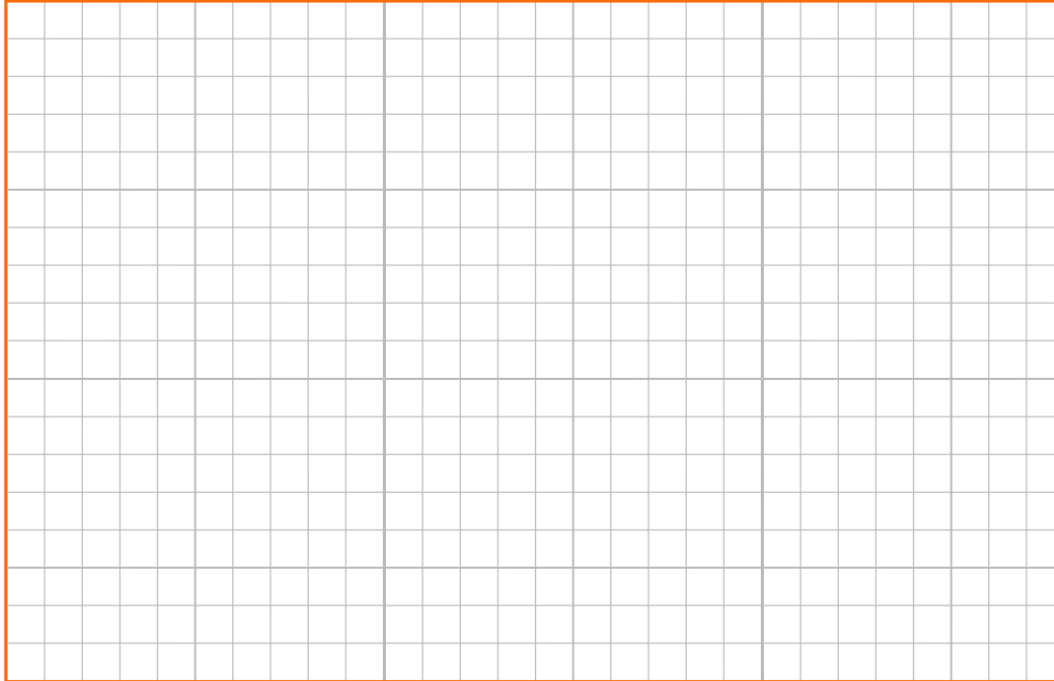


Diode U-I curves: Shockley curve parametrized with $U_T = 25.7$ mV, $I_S = 10^{-12}$ A. Characteristic curve of a 1N4001 Si-diode.

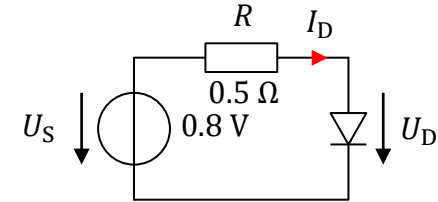
4.2.2 Diode modelling

Exercise: Simple diode circuit

You are given the following circuit. Please calculate I_D and U_D .



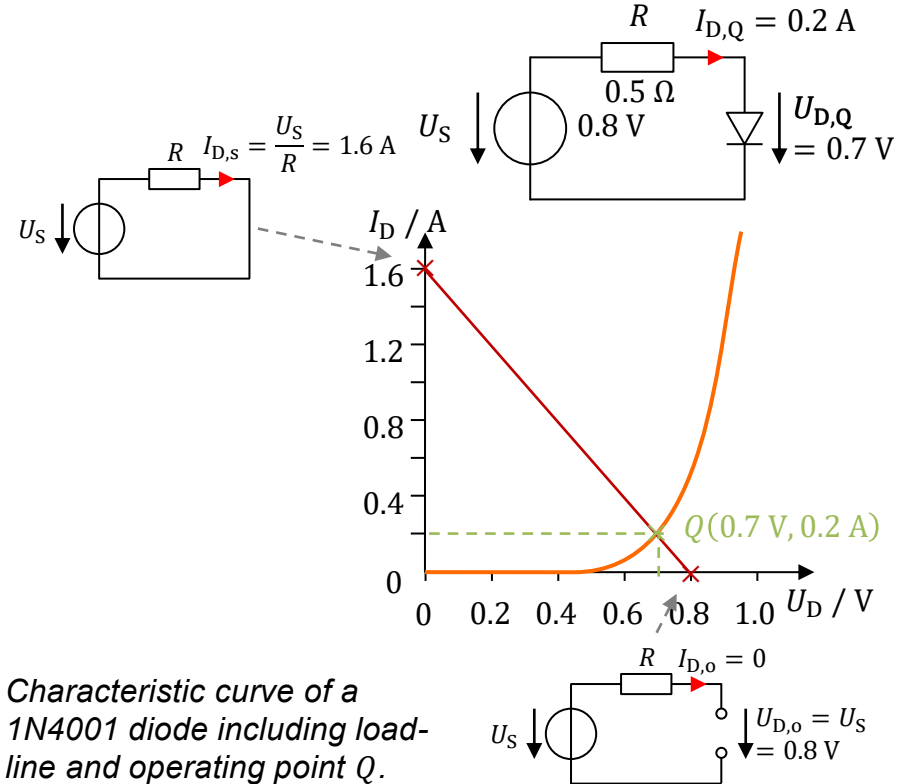
EXAM	Typical
•	Exam
•	Question
•	
•	



4.2.2 Diode modelling

load-line & operating point

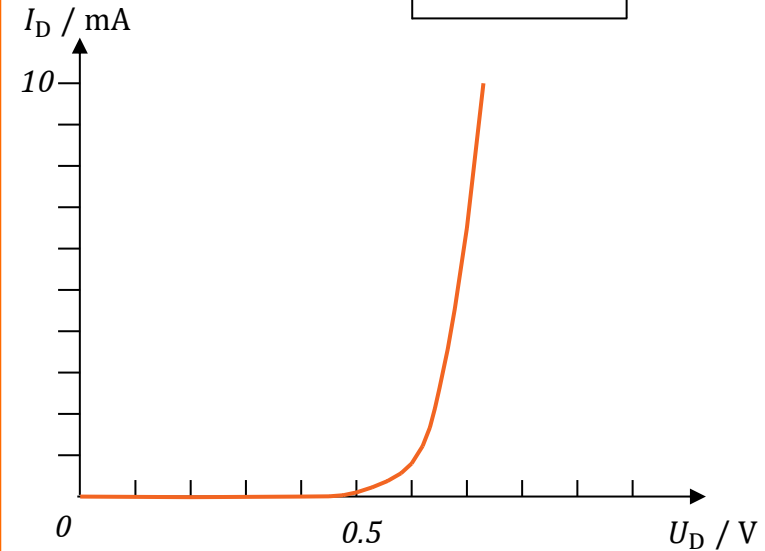
- A **load-line** is a **graphical method** to find the **operating point** $Q(U_{DQ}, I_{DQ})$ of a diode.
- **Procedure**
 - ⌊ Simplify the circuit.
 - ⌊ **Open circuit:** x -intercept at $I_{D,o} = 0$: $U_{D,o} = U_S$
 - ⌊ **Short circuit:** y -intercept at $U_{D,s} = 0$: $I_{D,s} = \frac{U_S}{R}$
 - ⌊ The **load-line** connects **both points** by a straight line.
 - ⌊ The Q-point is the **intercept** of the **load-line** with the diode's **characteristic curve**.
- Mathematical background:
 - ⌊ Kirchhoff's voltage law:
$$U_S = R I_D + U_D \quad \text{or} \quad I_D = -\frac{1}{R} U_D + \frac{U_S}{R}$$
 - ⌊ Where $I_D(U_D)$ describes a straight line:
$$y = m x + c$$
- **Note:** Also works with load resistor! (See Ex. 8.2)



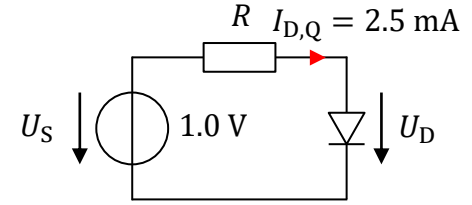
4.2.2 Diode modelling

Exercise: Si-diode circuit parametrization

Imagine the following circuit including a Si-diode. The voltage source ($U_S = 1.0\text{ V}$) drives a current $I_{D,Q} = 2.5\text{ mA}$ through the circuit. Please find the diode voltage $U_{D,Q}$ at the operating point and the resistance R .



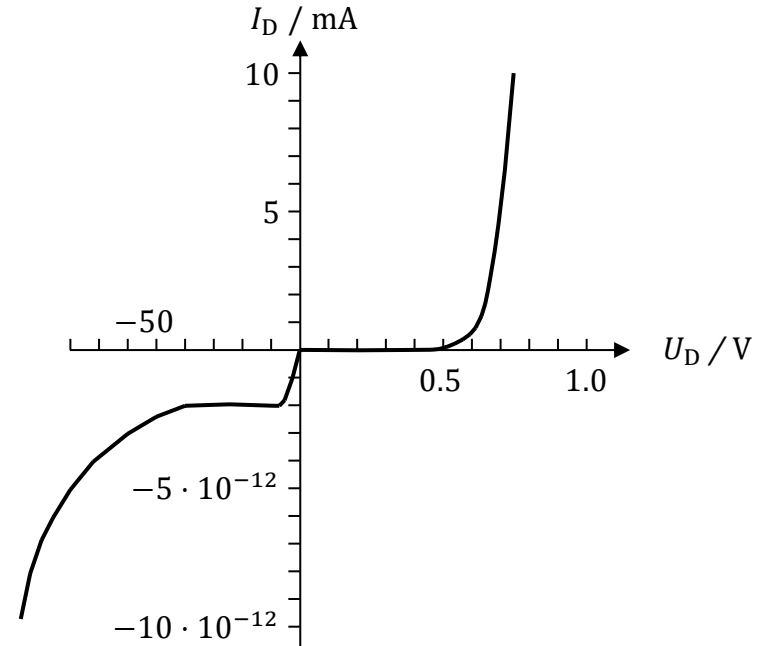
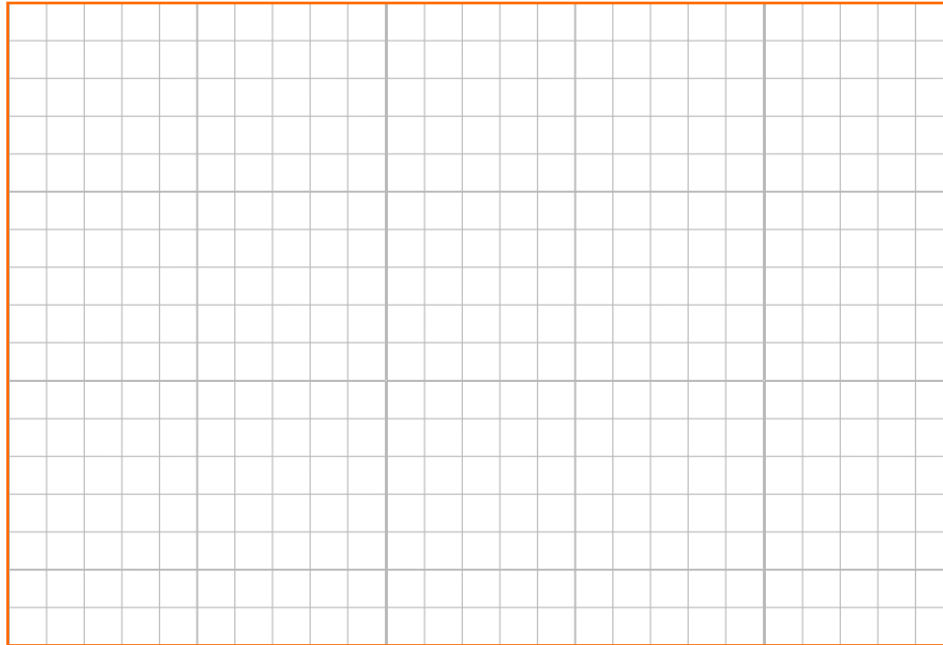
EXAM	Typical
•	Exam
•	Question
•	



4.2.2 Diode modelling

Exercise: 1N914 diode parametrization

Identify the reverse-bias saturation current I_S from the characteristic curve of a 1N914 Si-diode. Then calculate the diode voltage U_D for a forward current $I_D = 5 \text{ mA}$ at room temperature ($U_T = 25.7 \text{ mV}$). Calculate the differential resistance r_D via the small signal model. Compare the result with the graphic evaluation.



Recap

- Use the **ideal switch model** for qualitative rectifier behaviour.
- Use **switch + voltage source** for quick circuit estimates.
- Use **switch + voltage source + r_D** near a chosen operating point.
- Use the **diode equation** for more accurate current–voltage calculations.
- Use the **load-line** to find the operating point graphically.

4.2.2 Diode modelling

Chapter outline

LTspice simulation

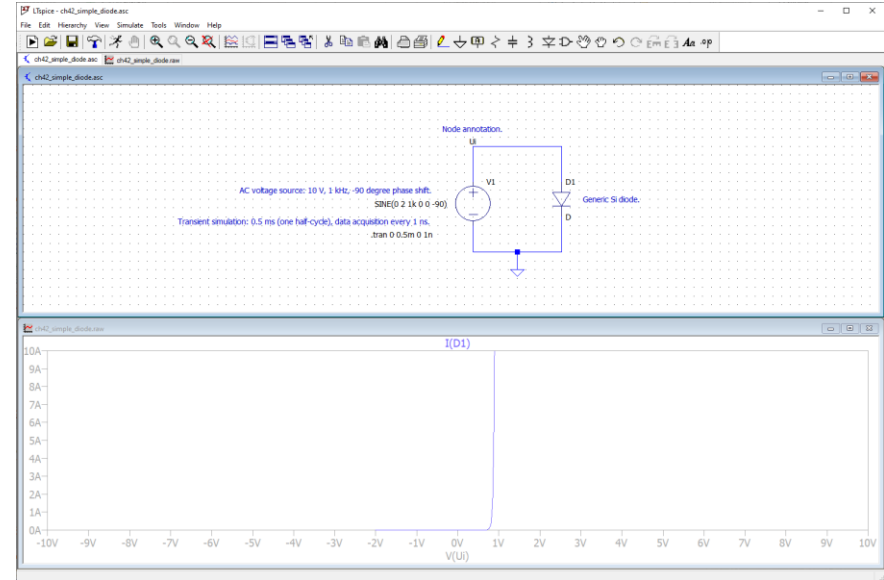
Experimental determination of r_D

Appendix to 4.2.2: Simulation and experiment

Appendix to 4.2.2: Simulation and experiment

LTspice

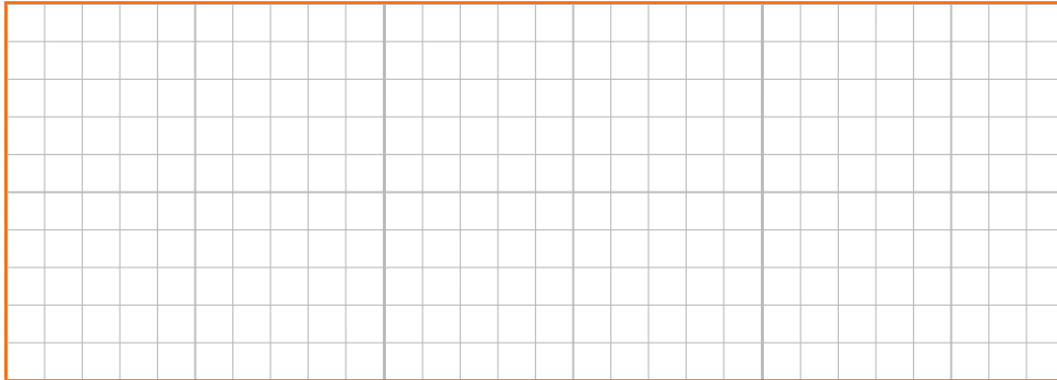
- LTspice® is a powerful, fast, and free **SPICE software**.
- **Simulate** virtually any **circuit** comprising **passive and active components**, e. g.
 - ⌈ Diode and transistor basic circuits.
 - ⌈ Diode and transistor characteristic curves.
- Verdict
 - ⌈ Steep BUT **quick learning curve**.
 - ⌈ Incredibly **powerful**.
 - ⌈ **Fast** circuit design.
 - ⌈ Extremely helpful for **visualizing simple and complex circuits**.
 - ⌈ **Highly recommended**.
- Download link of the current version
www.linear.com/LTspice



Appendix to 4.2.2: Simulation and experiment

Exercise: Si-diode characterization

Construct a circuit in LTspice comprising a standard Si diode in series with a voltage source. Use a triangular voltage sweep between -100 V and $+2\text{ V}$. What diode properties do you find?

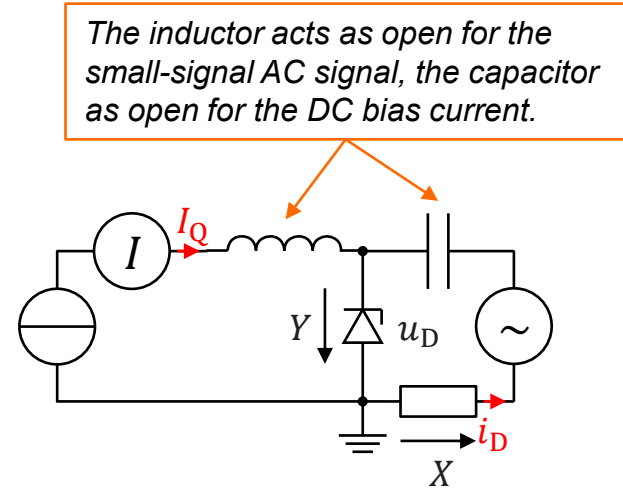


Appendix to 4.2.2: Simulation and experiment

Experimental determination of r_D

- The differential resistance of a diode around a certain operation can be measured via the superposition of an AC with a DC signal and an oscilloscope.
- 1. Set the operating point $Q(U_D, I_Q)$ via the constant current source.
- 2. Superimpose a small AC signal.
- 3. Plug the small-signal current i_D as X and the voltage u_D as Y -input of an oscilloscope.
- 4. The slope of the linear piece of the diode characteristic curve gives the differential resistance according to

$$r_D = \frac{dY}{dX} = \frac{du_D}{di_D} \approx \frac{\Delta u_D}{\Delta i_D}$$



Appendix to 4.2.2: Simulation and experiment

Exercise: Model-based determination of r_D

Built a schematic in LTSpice to determine the differential resistance of an arbitrary diode at 5 different operation points below and above threshold. *Hints: Use transient mode to simulate one full wave; Set the x-axis to the current i_R and the y-axis to the voltage u_D ; add two cursors to determine Δi_D , Δu_D and r_D from the slope; adjust the data compression for a smooth output as follows: .opt plotwinsize=0; .opt numdgt=12*

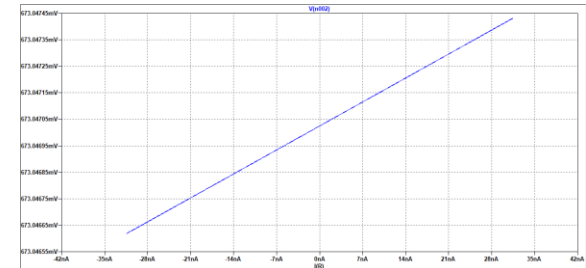
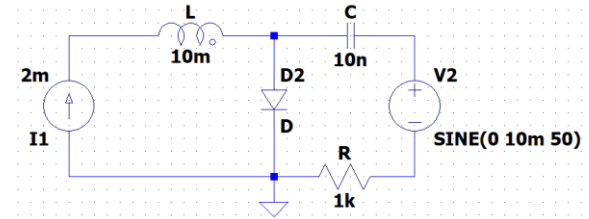


Waveform data compression settings

```
.opt plotwinsize=0  
.opt numdgt=12
```

Simulation params

```
.tran 0 40m 20m 1u
```



Chapter outline

- Rectifier circuits
- Voltage multiplier
- Voltage shifter
- Voltage regulator

4.2.3 Diode circuits

4.2.3 Diode circuits

Application: Single-phase rectifiers

— Half-wave rectification

- Input: single-phase sinusoidal voltage $u_i = \hat{U} \sin(\omega t)$.
- Output: Unidirectional, pulsating direct current $\rightarrow$ ripple noise.

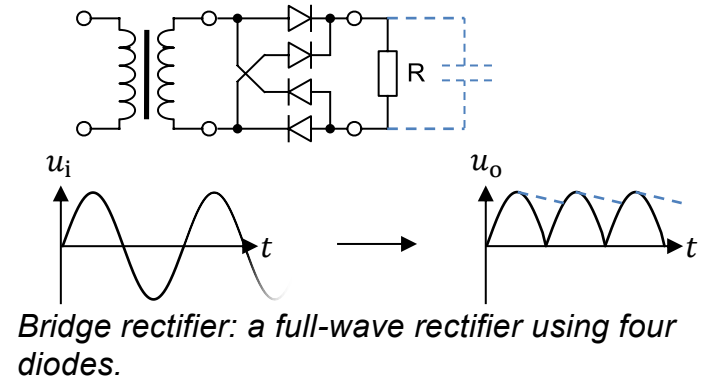
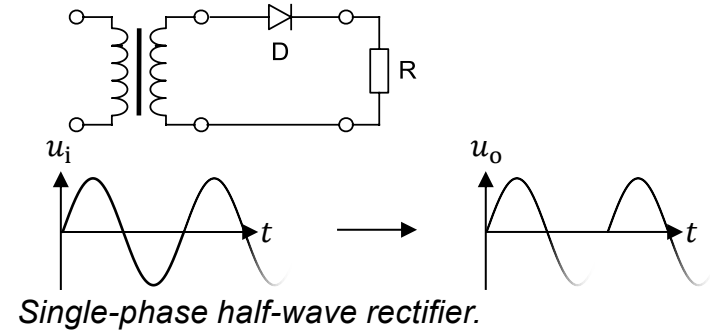
$$U_o = \frac{U_i}{\sqrt{2}} = \frac{\hat{U}}{2}$$

— Full-wave rectification

- Both polarities of input waveform are converted.
- Pulsating direct current w/ reduced ripple noise.

$$U_o = U_i = \frac{\hat{U}}{\sqrt{2}}$$

- Further improvement by adding a **capacitor in parallel**.
Slowly charging/discharging of the capacitor to reduce ripple voltage (*Brummspannung*).



4.2.3 Diode circuits

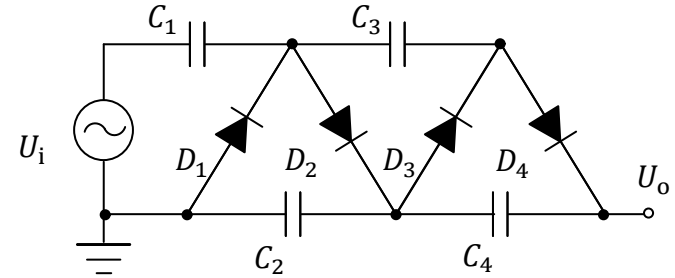
Application: Voltage multipliers

— Voltage multiplier

- ⌞ AC-to-DC converter
- ⌞ Half-wave or full-wave topologies
- ⌞ Each node multiplies input voltage by (almost) 2
- ⌞ Several cycles required to charge each capacitor
- ⌞ Each element only “sees” input voltage
- ⌞ Application in high voltage supplies

— Dickson multiplier / Dickson charge pump

- ⌞ (For details: ch42_Dickson_multiplier.asc in LTspice)
- ⌞ DC-to-DC converter
- ⌞ Operates with two clock pulses ϕ_1 and ϕ_2
- ⌞ Each clock cycle pushes the voltage of the capacitor plate to the next.
- ⌞ Small voltage only.
- ⌞ Application in integrated circuits (ICs)



A Cockcroft-Walton half-wave voltage quadrupler circuit.

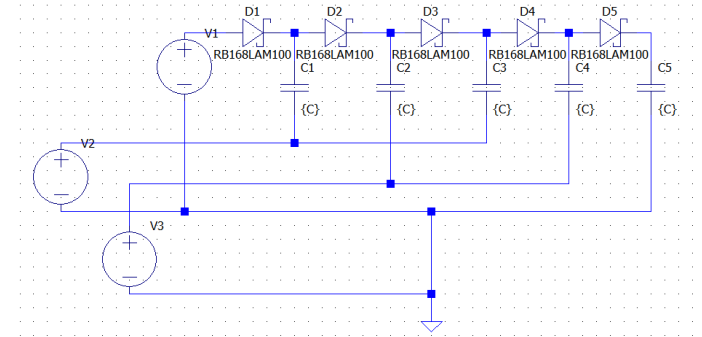
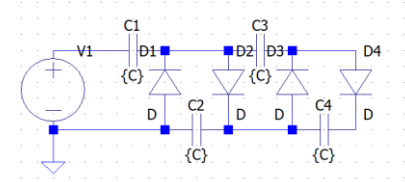
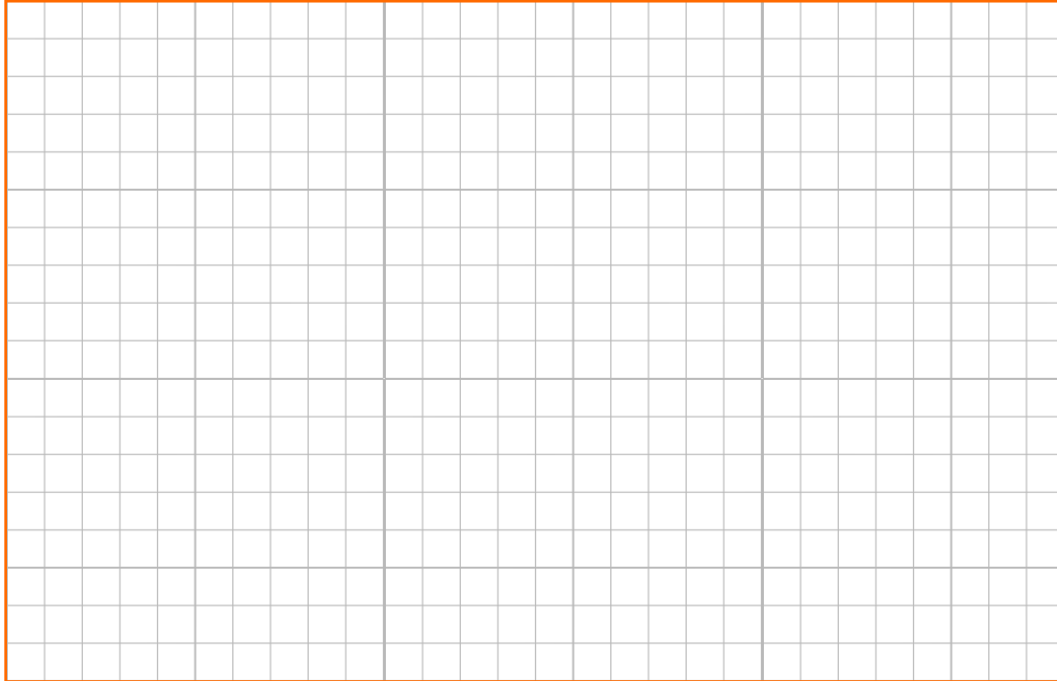


17 stage voltage multiplier module. It can produce up to 5.7 kV from 120 V AC.

4.2.3 Diode circuits

Homework: Voltage multiplier

Please analyse one of the two voltage multiplier, the Cockcroft-Walton or the Dickson multiplier, in LTSpice. Try understanding in which state the diodes are open or closed and what happens to the capacitors.



Top: Cockcroft-Walton multiplier.

Bottom: Dickson multiplier.

4.2.3 Diode circuits

Voltage shifter

- **Lowers the output voltage** by a quantity about equal to the diode's breakdown voltage.
- Typically implemented using a **Zener diode**
- Operating principle

1. For $U_i < U_Z$: Zener diode is non-conducting (open circuit)

$$U_o = 0$$

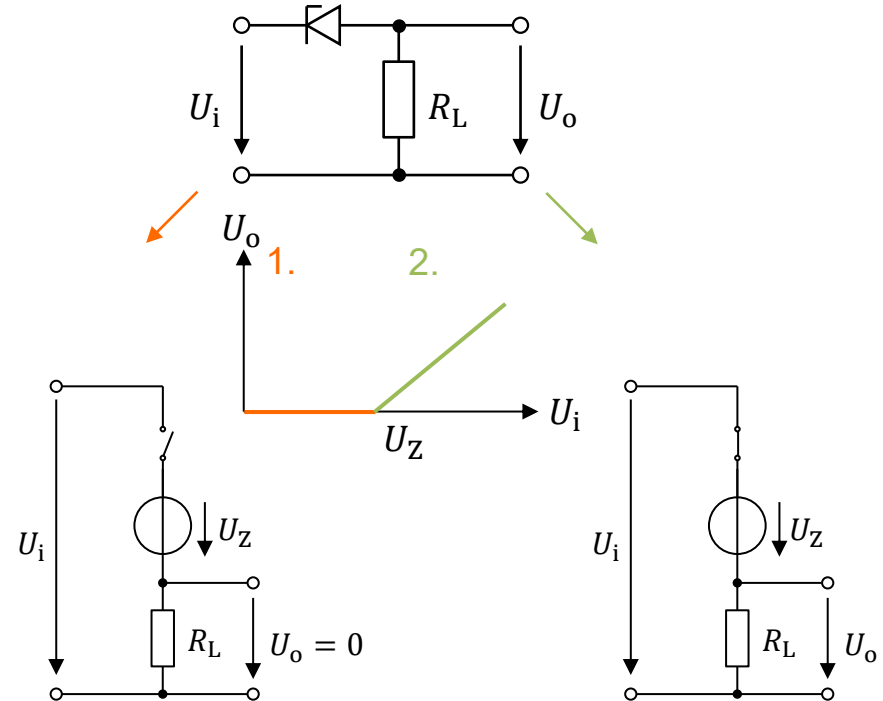
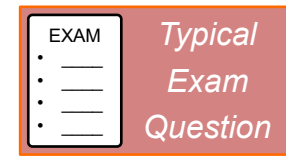
2. **Above Zener voltage** U_Z the diode becomes conductive.

⌈ Calculation either via voltage divider or KVL

$$U_o = \frac{R_L}{R_L + r_D} (U_i - U_Z) = U_i - U_Z - r_D \Delta I_D$$

⌈ In general, assume an ideal diode $r_D = 0$:

$$U_o = U_i - U_Z$$



Operating principle of a voltage shifter circuit for $r_D = 0$.

Case 1. $U_i < U_Z$ and case 2. $U_i \geq U_Z$

4.2.3 Diode circuits

Voltage regulator

- Limits the output voltage to the Zener voltage U_Z .
- If $U_i < \frac{R_L + R_S}{R_L} U_Z$ ("open switch"): R_L and R_S form a voltage divider.

Voltage divider solved for U_i

$$U_o = \frac{R_L}{R_L + R_S} U_i$$

- If $U_i \geq \frac{R_L + R_S}{R_L} U_Z$ ("switch closed"): R_L parallel to U_Z .

$$U_o = U_Z + r_D \Delta I_D$$

For an ideal diode i. e. $r_D = 0$:

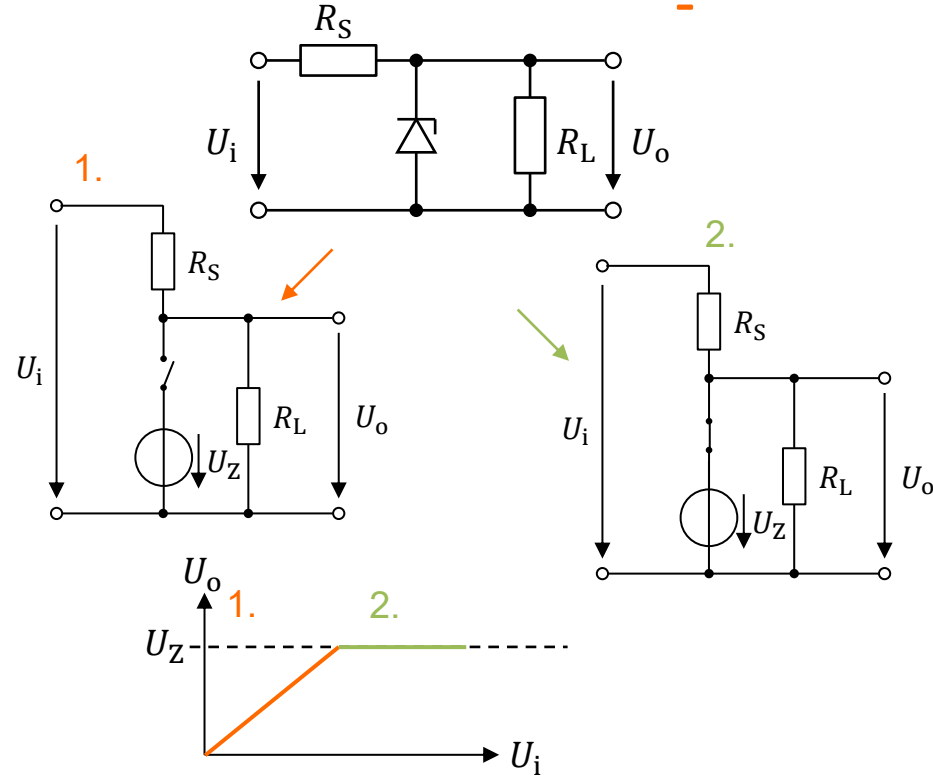
$$U_o = U_Z$$

- 7 The current through R_S divides according to:

$$I_{R_S} = I_Z + I_L = \frac{U_i - U_o}{R_S}$$

- **Warning:** Under open load the total current passes the diode and the power is maximised.

$$P_{Z \max} = I_{R_S} U_Z$$



Operating principle of a voltage shifter circuit for $r_D = 0$.

Case 1. $U_i < U_Z$ and case 2. $U_i \geq U_Z$

4.2.3 Diode circuits

Exercise: voltage regulator 1/2

EXAM	Typical
•	Exam
•	Question
•	
•	

You are given two voltage regulator circuits comprising lead acid battery with the input voltage $U_i = 12.6 \text{ V}$ series resistor $R_s = 10 \Omega$ and a Zener diode $U_Z = 5.6 \text{ V}$ in parallel to a load resistor $R_{L1} = 10 \Omega$ and $R_{L2} = 1 \text{ k}\Omega$, respectively. Calculate I_{R_s} , I_{R_L} and I_Z of both circuits. What do you notice?



4.2.3 Diode circuits

Exercise: voltage regulator 2/2

EXAM	Typical
• —	Exam
• —	Question
• —	
• —	

You are given two voltage regulator circuits comprising an input voltage $U_i = 12.6 \text{ V}$ series resistor $R_s = 10 \Omega$ and a Zener diode $U_Z = 5.6 \text{ V}$ in parallel to a load resistor $R_{L1} = 10 \Omega$ and $R_{L2} = 1 \text{ k}\Omega$, respectively. Determine I_Z of both circuits using the load-line method.



4.2.3 Diode circuits

Exercise

Calculate the output voltage U_o of a BZX79C5V6 Zener diode in a voltage regulated circuit at room temperature with load resistance $R_L = 50 \Omega$, a series resistance $R_S = 30 \Omega$ driven by an input voltage $U_i = 12 \text{ V}$. Is the operation compliant?



EXAM

•
•
•
•

Typical
Exam
Question



 TAIWAN SEMICONDUCTOR

BZX79C2V0 - BZX79C75
Taiwan Semiconductor

500mW, 5% Tolerance Zener Diodes

FEATURES

- Wide zener voltage range selection: 2.0V to 75V
- V_Z tolerance selection of $\pm 5\%$
- Compliant to RoHS directive 2011/65/EU and in accordance to WEEE 2002/96/EC
- Halogen-free according to IEC 61249-2-21

KEY PARAMETERS

PARAMETER	VALUE	UNIT
V_Z	2.0-75	V
Test current I_{ZT}	5	mA
P_D	500	mW
V_F at $I_F=100\text{mA}$	1.5	V
T_J Max.	175	$^{\circ}\text{C}$
Package	DO-35	
Configuration	Single die	

APPLICATIONS

- Low voltage stabilizers or voltage references
- Adapters
- Lighting application
- On-board DC/DC converter

MECHANICAL DATA



 TAIWAN SEMICONDUCTOR

BZX79C2V0 - BZX79C75
Taiwan Semiconductor

ELECTRICAL SPECIFICATIONS ($T_A = 25^{\circ}\text{C}$ unless otherwise noted)

PART NUMBER	MARKING CODE	ZENER VOLTAGE			TEST CURRENT	REGULAR IMPEDANCE		TEST CURRENT	LEAKAGE CURRENT	
		$V_Z @ I_{ZT}$			I_{ZT}	$Z_{ZT} @ I_{ZT}$	$Z_{ZK} @ I_{ZK}$	I_{ZK}	$I_{LK} @ V_{LK}$	
		V			mA	Ω	Ω	mA	μA	V
		Min.	Nom.	Max.		Max.	Max.		Max.	
BZX79C2V0	BZX79C2V0	1.88	2.0	2.12	5	100	600	1.0	150	1.0
BZX79C2V2	BZX79C2V2	2.08	2.2	2.33	5	100	600	1.0	150	1.0
BZX79C2V4	BZX79C2V4	2.28	2.4	2.56	5	100	600	1.0	100	1.0
BZX79C2V7	BZX79C2V7	2.51	2.7	2.89	5	100	600	1.0	75	1.0
BZX79C3V0	BZX79C3V0	2.8	3.0	3.2	5	95	600	1.0	50	1.0
BZX79C3V3	BZX79C3V3	3.1	3.3	3.5	5	95	600	1.0	25	1.0
BZX79C3V6	BZX79C3V6	3.4	3.6	3.8	5	90	600	1.0	15	1.0
BZX79C3V9	BZX79C3V9	3.7	3.9	4.1	5	90	600	1.0	10	1.0
BZX79C4V3	BZX79C4V3	4.0	4.3	4.6	5	90	600	1.0	5	1.0
BZX79C4V7	BZX79C4V7	4.4	4.7	5.0	5	80	500	1.0	3	2.0
BZX79C5V1	BZX79C5V1	4.8	5.1	5.4	5	60	480	1.0	2	2.0
BZX79C5V6	BZX79C5V6	5.2	5.6	6.0	5	40	400	1.0	1	2.0
BZX79C6V0	BZX79C6V0	5.6	6.0	6.4	5	30	300	1.0	0.5	2.0

Summary

- Diodes have a **non-linear, directional** current–voltage characteristic: conducting in forward bias $U_D > 0$, blocking in reverse bias $U_D < 0$.
- Si diodes are general-purpose; Zener diodes provide controlled reverse breakdown; Schottky diodes switch fast and have low forward voltage.
- Important quantities are threshold voltage U_{th} , breakdown voltage U_{br} , reverse saturation current I_S and the differential resistance r_D .
- A diode can be modelled by increasingly accurate models: **switch**, **switch + voltage source**, **switch + voltage source + r_D** , or the **diode equation**.
- Around a chosen operating point, the diode behaves like a resistor r_D for small AC signals.
- The **load-line method** is used to find the operating point Q .
- Typical applications are rectifiers, voltage shifters, and voltage regulators.

4.2 Semiconductor diodes

Version 4.0.0

- Initial version SoSe 26

Version 4.1.0

- Moved slide on Fermi distribution from appendix to 4.1.1, error correction, clarified slide on mobility

Version 4.2.0

- Minor structural changes and error correction.

Version 4.3.0

- Minor editorial updates, exercise added to chapter

Version 4.4.0

- Hint for exam relevant questions added